



UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL
FACULDADE DE MEDICINA
PROGRAMA DE PÓS-GRADUAÇÃO EM SAÚDE DA CRIANÇA E DO
ADOLESCENTE

CECÍLIA ROTAVA BURATTI

**Abordagem Integrada das Disfunções Respiratórias Agudas em Terapia Intensiva
Pediátrica: da Mecânica Respiratória à Segurança Ocupacional em Cenários de Alta
Transmissibilidade Viral**

Porto Alegre

2026

CECÍLIA ROTAVA BURATTI

**Abordagem Integrada das Disfunções Respiratórias Agudas em Terapia Intensiva
Pediátrica: da Mecânica Respiratória à Segurança Ocupacional em Cenários de Alta
Transmissibilidade Viral**

Tese apresentada ao Programa de Pós-Graduação em Saúde da Criança e do Adolescente da Faculdade de Medicina da Universidade Federal do Rio Grande do Sul como requisito parcial para a obtenção do título de doutorado em pediatria.

Orientador: Dr. Jefferson Pedro Piva

Porto Alegre

2026

CIP - Catalogação na Publicação

Rotava Buratti, Cecília
Abordagem Integrada das Disfunções Respiratórias
Agudas em Terapia Intensiva Pediátrica: da Mecânica
Respiratória à Segurança Ocupacional em Cenários de
Alta Transmissibilidade Viral / Cecília Rotava
Buratti. -- 2026.
69 f.
Orientador: Jefferson Pedro Piva.

Tese (Doutorado) -- Universidade Federal do Rio
Grande do Sul, Faculdade de Medicina, Programa de
Pós-Graduação em Saúde da Criança e do Adolescente,
Porto Alegre, BR-RS, 2026.

1. Terapia intensiva pediátrica. 2. Ventilação
mecânica. 3. Mecânica respiratória. 4. Biossegurança.
5. COVID-19. I. Piva, Jefferson Pedro, orient. II.
Título.

CECÍLIA ROTAVA BURATTI

**Abordagem Integrada das Disfunções Respiratórias Agudas em Terapia Intensiva
Pediátrica: da Mecânica Respiratória à Segurança Ocupacional em Cenários de Alta
Transmissibilidade Viral**

Tese apresentada ao Programa de Pós-Graduação em Saúde da Criança e do Adolescente da Universidade Federal do Rio Grande do Sul como requisito parcial para a obtenção do título de doutorado em pediatria.

Aprovado em: 03 de fevereiro de 2026.

BANCA EXAMINADORA

Doutor José Coletti Júnior
HOSPITAL ALBERT EINSTEIN

Doutora Rita de Cássia Silveira
HCPA/UFRGS

Doutora Cinara Andreolio
HCPA

*Dedico este trabalho a todos os
pacientes pediátricos e às suas famílias,
reafirmando meu compromisso com um
cuidado alicerçado em bases científicas
sólidas e permeado por humanidade/afeto.*

AGRADECIMENTOS

Aos meus dois grandes amores: meu companheiro de jornada, Gustavo, pela parceria, compreensão e constante incentivo; e ao Lorenzo, minha fonte inesgotável de amor e determinação.

À minha família — Diva, Valcir, Marcelo, Luiza, Silvia e Toni — por estarem sempre ao meu lado, doando afeto, carinho e tempo, para que meus sonhos se tornassem realidade.

Aos meus amigos da Terapia Intensiva Pediátrica, do antigo grupo do Hospital Universitário de Canoas, do Hospital da Criança Santo Antônio e do Hospital de Clínicas de Porto Alegre. E aqueles da vida toda — Renata Silva Duarte Santos, Verônica Indicatti Fiamenghi e Gabriel Tesche Roman. Sinto-me privilegiada por tantas oportunidades, trocas e por poder caminhar ao lado de vocês.

Ao meu orientador, Professor Jefferson Pedro Piva, minha profunda gratidão por acreditar em mim desde “pequena”, por caminhar junto e confiar-me voos cada vez mais ousados. És um mentor inspirador e generoso.

Ao Dr. Jouvett, que me acolheu com paciência e gentileza. Agradeço por confiar, ensinar, apoiar e oferecer inúmeras oportunidades. Tua presença foi verdadeiramente transformadora na minha jornada na Terapia Intensiva.

Ao Programa de Pós-graduação em Saúde da Criança e do Adolescente da UFRGS, pelo apoio e incentivo desde o surgimento da oportunidade do Fellowship em Pesquisa até o término deste doutorado.

Por fim, agradeço a todas as pessoas que, de forma direta ou indireta, contribuíram para a realização destes projetos de pesquisa e desta tese, e que, involuntariamente, deixaram de ser nominados.

RESUMO

Introdução: As disfunções respiratórias agudas figuram entre as principais causas de internação em Unidades de Terapia Intensiva Pediátrica (UTIP), correspondendo a mais de um terço das admissões e configurando-se como a forma mais prevalente de falência orgânica em crianças criticamente enfermas. O manejo dessas condições requer domínio da mecânica respiratória e rigor na adoção de medidas de biossegurança, especialmente em contextos de alta transmissibilidade viral, como durante a pandemia de COVID-19. **Objetivos:** Integrar os princípios da mecânica respiratória e de biossegurança no manejo de disfunções respiratórias agudas em UTIP, buscando estratégias que otimizem a ventilação mecânica e reduzam o risco ocupacional para a equipe multiprofissional. **Métodos:** Esta tese foi composta por dois estudos complementares. O primeiro, de caráter observacional retrospectivo, foi conduzido na UTIP do Hospital de Clínicas de Porto Alegre (Brasil) e incluiu 37 lactentes, de 1 a 24 meses, com insuficiência respiratória associada a infecção viral grave e submetidos à ventilação mecânica invasiva. O objetivo foi avaliar a relação entre a pressão inspiratória de pico (PIP) e a pressão de platô (P_{PLAT}), bem como a precisão de um modelo preditivo para estimativa de P_{PLAT} a partir da PIP. O segundo estudo, prospectivo e de coorte, foi realizado no Hospital Sainte-Justine (Canadá) e avaliou a eficácia do dispositivo *SplashGuard Caregiver* (SGCG) na redução da dispersão de aerossóis durante procedimentos geradores de aerossóis (PGA) realizados em pacientes com COVID-19. A análise envolveu a detecção de partículas virais no ambiente e na face de profissionais expostos a situações de alta transmissibilidade viral, sendo complementado por simulações de dinâmica de fluidos computacional (CFD - *computational fluid dynamics*) para analisar o comportamento aerodinâmico destas partículas. **Resultados:** No primeiro estudo, observou-se que a PIP superestimou significativamente P_{PLAT} , com diferença média de $7,3 \pm 4$ cmH₂O ($p < 0,01$), evidenciando as limitações de modelos preditivos baseadas apenas na PIP em lactentes com elevada resistência de vias aéreas. No segundo, a detecção de partículas virais reduziu de 5,9% na ausência do SGCG para 3% com o uso do dispositivo. As simulações CFD demonstraram redução de até 70% na concentração de aerossóis suspensos no ar quando mantidas aberturas mínimas no SGCG e que o posicionamento dos profissionais de saúde no sentido oposto ao principal fluxo de ar do ambiente pode promover redução adicional da exposição. **Conclusões:** A integração entre os princípios da mecânica respiratória e da biossegurança fortalece a qualidade da assistência em UTIP, favorecendo o desenvolvimento de estratégias ventilatórias mais seguras e individualizadas, além de contribuir para a mitigação dos riscos ocupacionais.

Palavras-chave: terapia intensiva pediátrica; ventilação mecânica; mecânica respiratória; biossegurança; COVID-19.

ABSTRACT

Introduction: Acute respiratory dysfunctions are among the leading causes of admission to Pediatric Intensive Care Unit (PICU), accounting for more than one third of hospitalizations and representing the most prevalent form of organ failure in critically ill children. The management of these conditions requires a solid understanding of respiratory mechanics and strict adherence to biosafety measures, particularly in settings of high viral transmissibility, such as during the COVID-19 pandemic. **Objective:** To integrate the principles of respiratory mechanics and biosafety in the management of acute respiratory dysfunctions in PICU, aiming to develop strategies that optimize mechanical ventilation and reduce occupational risk for the multidisciplinary healthcare team. **Method:** This thesis comprised two complementary studies. The first research, a retrospective observational study, was conducted in the PICU of Hospital de Clínicas de Porto Alegre (Brazil) and included 37 infants, aged 1 to 24 months, with respiratory failure associated with severe viral infection who required invasive mechanical ventilation. The objective was to assess the relationship between peak inspiratory pressure (PIP) and plateau pressure (P_{PLAT}), as well as to evaluate the accuracy of a predictive model for estimating P_{PLAT} from PIP values. The second study, a prospective cohort conducted at Sainte-Justine Hospital (Canada), evaluated the efficacy of the *SplashGuard Caregiver* (SGCG) device in reducing aerosol dispersion during aerosol-generating procedures (AGP) performed on patients with COVID-19. The analysis involved detecting viral particles in the environment and on the faces of healthcare workers exposed to high-transmissibility settings, complemented by computational fluid dynamics (CFD) simulations to examine the aerodynamic behavior of these particles. **Results:** In the first study, PIP significantly overestimated P_{PLAT} , with a mean difference of 7.3 ± 4 cmH₂O ($p < 0.01$), highlighting the limitations of predictive models based solely on PIP in infants with high airway resistance. In the second study, viral particle detection decreased from 5.9% in the absence of the SGCG to 3% with device use. CFD simulations demonstrated up to a 70% reduction in the concentration of airborne aerosols when minimal openings were maintained in the SGCG and indicated that positioning healthcare professionals opposite to the main airflow direction of the environment could further reduce exposure. **Conclusions:** The integration of respiratory mechanics and biosafety principles enhances the quality of care in the PICU, promoting the development of safer and more individualized ventilatory strategies, while also contributing to the mitigation of occupational risks.

Keywords: pediatric intensive care; mechanical ventilation; respiratory mechanics; biosafety; COVID-19.

LISTA DE ILUSTRAÇÕES

Figura 1 - Características do protótipo final do SplashGuard Caregiver.....	20
Figura 2 - <i>Reliability of peak inspiratory pressure in decelerating-flow mode (PIP) to estimate plateau pressure during square-flow mode (P_{PLAT}). (A): Scatterplot of P_{PLAT} and PIP in the 37 subjects studied ($r = 0.57$, $p < 0.001$). (B): Bland-Altman analysis comparing P_{PLAT} and eP_{PLAT} calculated using the Ilia et al. equation, where the average difference between P_{PLAT} and eP_{PLAT} was 1.0 ± 4.0 mmHg with 95% limits of agreement of -6.9 and 8.8.....</i>	37
Figura 3 - <i>SplashGuard Caregiver prototype</i>	43
Figura 4 - <i>Intensive care room simulation. (A) Room model. (B) Individual without SplashGuard Caregiver (SGCG). (C) First configuration of individual with SGCG (140-mm gap). (D) Second configuration of individual with SGCG (20-mm gap)</i>	47
Figura 5 – <i>Simulation analysis. (A) Airflow leakage through the gap between the doors and the floor (two-dimensional map e velocity magnitude). (B) Airflow leakage inside the room and around the SplashGuard Caregiver (SGCG) [three-dimensional (3D) envelope e velocity magnitude >0.4 m/s]. (C) 3D visualization of the air flow entering the SGCG through the opening created by the plastic film (140- mm-gap scenario). (D) Air flow does not enter the SGCG through the reduced opening created by the plastic film (20-mm-gap scenario)</i>	49
Figura 6 - <i>Three-dimensional visualization of the numerically modelled airborne particles emitted by a patient and dispersed throughout the room. (A) In the baseline configuration at t 25 s. (B) In the first SplashGuard Caregiver (SGCG) configuration (140-mm gap) at t 25 s. (C) In the second SGCG configuration (20-mm gap) at t 25 s. (D) Virtual representation of the breathing zone (orange box) around the healthcare worker</i>	51
Figura 7 - <i>Evolution of the cumulative number of particles passing through the healthcare worker's breathing zone over time depending on the SplashGuard Caregiver (SGCG) configuration.....</i>	51
Figura 8 - <i>Breakdown of extracted, deposited and airborne particles within the room and the SplashGuard Caregiver (SGCG) for each configuration after 100 s of physical time</i>	52

LISTA DE TABELAS

Tabela 1 - Variáveis de mecânica pulmonar com respectivo detalhamento.....	17
Tabela 2 - <i>Patients' characteristics</i>	35
Tabela 3 - <i>Characteristics of the sample and comparison in prevalence of quantitative polymerase chain reaction (qPCR) samples between SplashGuard Caregiver (SGCG) and non-SGCG groups</i>	48

LISTA DE ABREVIATURAS E SIGLAS

AutoPEEP	<i>Positive-end expiratory pressure</i> gerado pelo paciente
CFD	Dinâmica de fluidos computacional (<i>computacional fluid dynamic</i>)
C ou C _{ST}	Complacência ou complacência estática
DP	<i>Driving pressure</i>
EPI	Equipamento de proteção individual
FR	Frequência respiratória
MP	<i>Mechanical power</i>
PALICC	<i>Pediatric Acute Lung Injury Consensus Conference</i>
PGA	Procedimentos geradores de aerossóis
PEEP	<i>Positive-end expiratory pressure</i>
PIP	<i>Positive inspiratory pressure</i>
P _{PLAT}	Pressão platô
R ou R _I /R _E	Resistência ou resistência inspiratória/expiratória
SDRA	Síndrome do desconforto respiratório agudo
SGCG	<i>SplashGuard Caregiver</i>
UTIP	Unidade de terapia intensiva pediátrica
VM	Ventilação mecânica ou ventilação mecânica invasiva
VNI	Ventilação mecânica não-invasiva
V _T	Volume corrente
VILI	Lesão induzida pela ventilação (<i>ventilator-induced lung injury</i>)

SUMÁRIO

1 INTRODUÇÃO	12
2 REVISÃO DA LITERATURA.....	14
3 JUSTIFICATIVA	22
4 OBJETIVOS	23
4.1 OBJETIVO GERAL.....	23
4.2 OBJETIVOS ESPECÍFICOS	23
5 MÉTODOS.....	24
6 RESULTADOS (ARTIGOS).....	28
7 CONCLUSÕES.....	58
8 CONSIDERAÇÕES FINAIS	59
REFERÊNCIAS.....	60
ANEXO A – CERTIFICADO RELATIVO AO FELLOWSHIP DE PESQUISA EM FISIOLOGIA DO SUPORTE VENTILATÓRIO EM CRIANÇAS	68
ANEXO A – <i>FRAMEWORK</i> CONCEITUAL: INTEGRANDO MECÂNICA RESPIRATÓRIA E BIOSSEGURANÇA	69

1 INTRODUÇÃO

As disfunções respiratórias agudas correspondem a mais de um terço das internações em unidade de terapia intensiva pediátrica (UTIP) e, desde 2010, representam o tipo mais prevalente de falência orgânica (YEHYA *et al.*, 2022; KILLIEN *et al.*, 2023). A ampla diversidade etiológica — que abrange principalmente as pneumonias virais, seguidas pelas bacterianas, pelas exacerbações de doenças pulmonares crônicas e pela síndrome do desconforto respiratório agudo (SDRA) — exige uma abordagem individualizada e um profundo entendimento da fisiopatologia respiratória (DE LUCA *et al.*, 2023; COHEN *et al.*, 2024; DREßEN *et al.*, 2025).

O manejo destas condições é norteado pelo suporte ventilatório, que inclui desde oxigenioterapia, a ventilação mecânica não-invasiva (VNI) e ventilação mecânica invasiva (VM). A gravidade do quadro associa-se a fatores de risco como prematuridade, doenças neuromusculares, imunossupressão, cardiopatias congênitas e coinfeções bacterianas. Pacientes com esses fatores de risco tendem a apresentar maior necessidade de VM e de tempo de internação, constituindo justamente os grupos com maior diversidade fisiopatológica, representando na prática um importante desafio terapêutico (O'BRYANT *et al.*, 2022; MAZUR; CABALLERO; NUNES, 2024). Neste cenário, a mecânica respiratória consolidou-se como uma ferramenta essencial para o entendimento do comportamento do sistema respiratório durante o curso de VM (PETERSON-CARMICHAEL *et al.*, 2016).

A análise de parâmetros como complacência (C), resistência (R), pressão platô (P_{PLAT}) e Auto-PEEP permite identificar padrões fisiopatológicos — restritivos ou obstrutivos — e individualizar a estratégia ventilatória. Essa avaliação é tida como chave para garantir uma estratégia de ventilação pulmonar protetora na SDRA, a qual vem sendo associada a redução de mortalidade e de risco de lesão pulmonar induzida pelo ventilador (*ventilator-induced lung injury* – VILI) (BHALLA *et al.*, 2021; EMERIAUD *et al.*, 2023; KNEYBER *et al.*, 2023; BEAUCHAMP; THÉRIAULT; SAUTHIER, 2025). Embora amplamente estudados em populações adultas, esses conceitos de mecânica respiratória carecem de consolidação e validação em Pediatria, dada a expressiva diferença anatômica, a maior complacência da caixa torácica e as demandas metabólicas específicas dessa faixa etária (TEPPER *et al.*, 2001; SEED; WILSON; COATES, 2012; DI CICCIO *et al.*, 2021).

Como já mencionado, os vírus desempenham um papel central nas disfunções respiratórias, e as discussões relacionadas à contaminação viral do paciente crítico são rotineiras. Mais recentemente, intensificou-se também a preocupação com o risco de

contaminação da equipe multiprofissional no cuidado destes pacientes. Esse debate ganhou destaque com o surgimento do COVID-19, especialmente em ambientes de cuidados intensivos, onde procedimentos geradores de aerossóis (PGA) são frequentes e o contato próximo com os pacientes é inevitável, elevando o risco ocupacional (TIAN *et al.*, 2022).

Durante a pandemia, observou-se a implementação rigorosa de equipamentos de proteção individual (EPI), revisão de protocolos de controle de infecção, treinamento intensivo das equipes, fortalecimento do monitoramento epidemiológico, com impactos significativos na redução deste risco (LEFEBVRE *et al.*, 2023). O *SplashGuard Caregiver* (SGCG) – uma caixa de acrílico instalada sobre a cabeça e o dorso dos pacientes com aberturas laterais conectadas a pressão negativa – surge como um dispositivo adicional para reduzir o risco de contaminação no contexto de PGA (FRANÇOIS *et al.*, 2020). A preocupação com biossegurança também se expandiu para outros vírus respiratórios, como o sincicial respiratório, o influenza e o rinovírus, cujo aumento da circulação após a pandemia tem sido acompanhado por elevada frequência de casos na UTIP (KARA *et al.*, 2024).

Esta tese propõe, portanto, uma abordagem integrada das disfunções respiratórias agudas em UTIP, articulando dois eixos complementares: 1) a utilização de medidas e conceitos derivados da mecânica respiratória para otimização de VM e individualização de tratamento; e 2) a adoção de estratégias que visem reduzir os riscos ocupacionais associados à assistência respiratória em ambientes de alta transmissibilidade viral. Seu propósito é contribuir para a consolidação de práticas assistenciais que unam rigor científico, segurança e humanização, fortalecendo o vínculo entre fisiologia, tecnologia e biossegurança. Para atingir esse objetivo, o estudo foi desenvolvido em etapas sucessivas, iniciando por uma ampla revisão de fisiopatologia/mecânica respiratória e riscos ocupacionais, seguidas pelo desenvolvimento de um estudo observacional em mecânica respiratória e de uma coorte prospectiva em biossegurança durante a pandemia de COVID-19.

2 REVISÃO DA LITERATURA

2.1 Disfunções Respiratórias Agudas

As disfunções respiratórias agudas estão entre as principais causas de internação e morbimortalidade em UTIP, representando mais de um terço das admissões e correspondendo ao tipo de falência orgânica mais prevalente em pacientes críticos (KILLIEN *et al.*, 2023). Englobam um espectro de condições clínicas caracterizadas por comprometimento da oxigenação e/ou ventilação, cuja definição considera os diferentes graus de hipoxemia e a necessidade de suporte ventilatório (YEHYA *et al.*, 2023). As principais etiologias incluem pneumonias virais, bacterianas ou aspirativas, exacerbações de doenças pulmonares crônicas, crises asmáticas e SDRA de causa pulmonar ou extrapulmonar (COHEN *et al.*, 2024; DREßEN *et al.*, 2025).

O padrão de maturação pulmonar e imunológica é dinâmico ao longo da infância, influenciando a resposta inflamatória e a suscetibilidade a agentes infecciosos, especialmente aos vírus, principais causas de insuficiência respiratória aguda em pediatria (KNEYBER *et al.*, 2023). Em crianças, principalmente lactantes, há predominância de respostas do tipo Th2, menor produção de interferons (especialmente IFN- γ e IFN- α), memória imunológica e capacidade de resposta adaptativa, tornando a imunidade inata o principal mecanismo de defesa inicial contra vírus respiratórios (LAMBERT; CULLEY, 2017; HEINONEN *et al.*, 2019; VERHOEVEN, 2019). A resposta inflamatória pode ser desregulada, com produção excessiva de citocinas pró-inflamatórias (IL-6, IL-1 β), menor eficiência na eliminação viral e ambiente imunológico tolerogênico, o que contribui para quadros mais graves e recorrentes de pneumonia viral, com risco aumentado de sobreposição bacteriana (COATES *et al.*, 2015; JONES *et al.*, 2019).

A presença de disfunções respiratórias agudas associa-se a piores desfechos, incluindo risco de evolução para SDRA, aumento do tempo de VM e a maior incidência de falência de múltiplos órgãos que corresponde ao principal fator independente de mortalidade em crianças com insuficiência respiratória aguda (LÓPEZ-FERNÁNDEZ *et al.*, 2012). A SDRA pediátrica representa uma entidade de elevada complexidade diagnóstica e terapêutica com incidência de apenas 3%, mas elevada taxa de mortalidade chegando a até 17% (KHEMANI *et al.*, 2019). Desde a definição do *Pediatric Acute Lung Injury Consensus Conference* (PALICC), em 2015, houve importante avanço na padronização de critérios diagnósticos, estratificação de gravidade e aplicação de estratégias ventilatórias protetoras adaptadas à população pediátrica (JOUVET

et al., 2015; EMERIAUD *et al.*, 2023). Entretanto, persiste uma lacuna entre o conhecimento teórico e a incorporação sistemática dessas práticas na rotina clínica, especialmente em contextos de alta demanda e recursos limitados (BHALLA *et al.*, 2021).

O manejo das disfunções respiratórias em pediatria deve considerar a diversidade anatômica e funcional das diferentes faixas etárias, que influenciam diretamente a mecânica respiratória e a resposta à VM (TRACHSEL *et al.*, 2022; KNEYBER *et al.*, 2023). A maior complacência da parede torácica, o calibre reduzido das vias aéreas, a menor reserva funcional e o consumo metabólico elevado tornam o sistema respiratório pediátrico particularmente vulnerável à fadiga muscular e à hipoxemia sustentada (DI CICCIO *et al.*, 2021).

2.2 Repercussão das Disfunções Respiratórias no Paciente Pediátrico – Mecânica Respiratória

A mecânica respiratória fundamenta-se na interação entre pressão, volume e fluxo no sistema respiratório, sendo regida por propriedades físicas que incluem complacência, elastância, resistência e trabalho respiratório (KAMINSKY; COCKCROFT; DAVIS, 2023). Esses elementos constituem a base fisiológica para a compreensão tanto da respiração espontânea, em que o gradiente de pressão é gerado pela ação dos músculos respiratórios, quanto da ventilação mecânica, na qual a pressão positiva aplicada pelo ventilador modifica os determinantes da distensão pulmonar e da movimentação de ar (GRINNAN; TRUWIT, 2005).

A avaliação da mecânica respiratória pode ser realizada por métodos como espirometria, oscilometria, técnicas de interrupção do fluxo e manometria esofágica, sendo que a escolha do método depende da idade e da capacidade de cooperação da criança (ARETS; VAN DER ENT, 2004; BOAS; REN, 2025). Em pacientes críticos, a análise da mecânica respiratória é feita através do próprio ventilador mecânico e é fundamental para ajustar estratégias ventilatórias, monitorar a evolução clínica e minimizar lesão pulmonar induzida pela ventilação (HESS, 2014; PETERSON-CARMICHAEL *et al.*, 2016; ANDREOLIO *et al.*, 2021; CRUCES, 2023). A **Tabela 1** resume as principais variáveis de mecânica respiratória que permitem identificar padrões fisiopatológicos (restritivos e obstrutivos) e individualizar a estratégia ventilatória.

O conceito de lesão pulmonar induzida pela ventilação (VILI, *ventilator-induced lung injury*) engloba diferentes mecanismos de dano, incluindo barotrauma, volutrauma, atelectrauma, biotrauma e ergotrauma, e tem sido amplamente discutido nos últimos anos. Estratégias de ventilação protetora – que derivam diretamente dos princípios da mecânica respiratória – voltadas à limitação do *stress* (variação de pressão) e do *strain* (deformação

pulmonar representada pelo volume corrente) – buscando evitar tanto o colapso alveolar quanto hiperdistensão, constituem pilares fundamentais para a redução da VILI (GATTINONI *et al.*, 2017). Entre as práticas recomendadas na VM protetora em SDRA tem-se: limitação de volumes correntes (V_T), pressão platô (P_{PLAT}), *Driving Pressure* (DP) e titulação de pressão expiratória final positiva (PEEP) de acordo com FiO_2 considerando as melhores respostas de complacência estática do sistema respiratório (C_{ST}) e da DP (EMERIAUD *et al.*, 2023).

O uso rotineiro de medidas de mecânica respiratória na UTIP permanece limitado por dificuldades técnicas, pela grande variabilidade dos tamanhos corporais e pela ausência de conceitos consolidados e validados especificamente para a população pediátrica (PETERSON-CARMICHAEL *et al.*, 2016). Além disso, a maioria dos estudos de mecânica respiratória conduzidos em adultos, foi realizada em pacientes ventilados em modo volume controlado, na ausência de esforço respiratório espontâneo e de hiperinsuflação dinâmica, condições necessárias para a obtenção adequada de P_{PLAT} e C_{ST} , enquanto a titulação de PEEP se deu em modo pressão controlada (HESS, 2014; AMATO *et al.*, 2015).

No contexto pediátrico, observa-se predominância do uso de modos ventilatórios de fluxo desaceleração, ou seja, pressão controlada e pressão regulada volume controlado, nos quais a P_{PLAT} é raramente medida, sendo frequentemente substituída por pressão inspiratória de pico (PIP) (PANICO *et al.*, 2015; BHALLA *et al.*, 2021). Embora alguns estudos tenham demonstrado boa correlação entre PIP e P_{PLAT} em crianças com doença pulmonar restritiva e mediana de idade de 8,6 anos, a realidade da UTIP inclui um grande contingente de lactentes com importante componente obstrutivo, condição em que essa correlação tende a ser menos precisa (BURATTI; PIVA; JOUVET, 2022; PATEL; THOMAS; YEHYA, 2022).

Tabela 1: Variáveis de mecânica pulmonar com respectivo detalhamento.

Variável	Definição	Fórmula	Valores de referência	Aferição	Relevância Clínica
Pressão de Platô (P_{PLAT})	Reflete a pressão aplicada nas pequenas vias aéreas e nos alvéolos ao final da inspiração		≤ 28 cmH ₂ O ou ≤ 32 cmH ₂ O em paciente com redução de complacência de caixa torácica (EMERIAUD <i>et al.</i> , 2023)	Através da pausa inspiratória (0,5 seg) em VC ou em PC/PRVC quando em fluxo zero (PETERSON-CARMICHAEL <i>et al.</i> , 2016)	Valores elevados estão associados a mais risco de VILI ou sobredistensão alveolar (CRUCES <i>et al.</i> , 2023)
Complacência do Sistema Respiratório (C_{SR})	Capacidade de distensão pulmão/caixa torácica; oposto da elastância	$C_{DYN} = V_T / (PIP - PEEP)$ $C_{ST} = V_T / (P_{PLAT} - PEEP)$	Adultos: 60–100 cmH ₂ O Pediatria: 1 ml/kg/cmH ₂ O SDRA < 0,6 ml/kg/cmH ₂ O BVA grave ~ 0,55 ml/kg/H ₂ O (ANDREOLIO <i>et al.</i> , 2021; DUARTE <i>et al.</i> , 2025)	C _{DYN} é medida continuamente pelo ventilar e C _{ST} é aferida após pausa inspiratória (2 - 5 seg) (PETERSON-CARMICHAEL <i>et al.</i> , 2016)	Identificar precocemente deteriorações clínicas por atelectasia, edema, sobredistensão e titular PEEP em SDRA (KASKINEN <i>et al.</i> , 2018; KNEYBER <i>et al.</i> , 2023)
Resistência (R)	Oposição ao fluxo, dependente do calibre das vias aéreas e viscosidade do gás	$R = PIP - P_{PLAT} / Fluxo$	PMT: 80; RN: 40; 1 ano: 15 e 7 anos: 4 L/seg, variando com o peso e podendo chegar de 40 – 140 L/seg em BVA grave (LÓPEZ-HERCE CID, 2003; ANDREOLIO <i>et al.</i> , 2021; BRUNO <i>et al.</i> , 2022)	Pausa inspiratória para determinar R inspiratória e pausa expiratória para determinar R expiratória (2 - 5 seg) (PETERSON-CARMICHAEL <i>et al.</i> , 2016)	Aumentos podem indicar broncoespasmo, secreção ou edema (BRUNO <i>et al.</i> , 2022)
Auto-PEEP	Hiperinsuflação dinâmica por esvaziamento incompleto	Auto-PEEP = PEEP total – PEEP extrínseca (ventilador)	Não existe um valor de referência. BVA: 1 – 5 cmH ₂ O (ANDREOLIO <i>et al.</i> , 2021)	Pausa expiratória (2 – 5 seg) com ausência de esforço (HESS, 2014)	Ao quantificar o alçaponamento de ar, permite otimizar estratégias de VM
Driving Pressure (DP)	Pressão que representa o tamanho funcional do pulmão	$DP = P_{PLAT} - PEEP$ ou $DP = V_T / C_{SR}$ (AMATO <i>et al.</i> , 2015)	≤15 cmH ₂ O (AOYAMA <i>et al.</i> , 2018; VAN SCHELVEN <i>et al.</i> , 2021).	Medir P _{PLAT} com pausa inspiratória (0,5 seg) e calcular diferença para PEEP	DP ≥ 15 cmH ₂ O associa-se a mortalidade e discrimina SDRA em pediatria (DÍAZ <i>et al.</i> , 2021; VAN SCHELVEN <i>et al.</i> , 2021)
Mechanical Power (MP)	Energia mecânica/min aplicada ao sistema respiratório	$MP = 0.098 \times FR \times V_T \times (PIP - \frac{1}{2} DP)$	Adultos: >17 J/min ou < 12 J/min (nada definitivo) (GUÉRIN <i>et al.</i> , 2016; SERPA NETO <i>et al.</i> , 2018)	Descrito acima	MP elevado está associado a mortalidade em Pediatria (PERCY <i>et al.</i> , 2023)
Constante de Tempo	Tempo necessário para enchimento/esvaziamento alveolar	$\tau = R \times C$	Dependente de R e C; maior em doenças obstrutivas	Descrito acima	Permite identificar risco de auto-PEEP, hiperinsuflação dinâmica e ventilação ineficaz, além de orientar o ajuste do tempo ins/expiratório e da frequência respiratória

Fonte: Elaboração própria.

2.3 Repercussão das disfunções respiratórias na equipe assistencial

2.3.1 Riscos ocupacionais na assistência respiratória

O ambiente da UTIP é caracterizado pela exposição da equipe multiprofissional a riscos ocupacionais elevados, especialmente em períodos de alta transmissibilidade viral — como durante a pandemia de COVID-19, surtos de influenza, vírus sincicial respiratório e outros patógenos respiratórios (BANSAL *et al.*, 2025). A Organização Mundial da Saúde estimou que profissionais da saúde representaram cerca de 11% dos casos notificados de COVID-19, com variações entre países de acordo com a capacidade de testagem. A taxa de contaminação foi maior entre aqueles que atuavam em terapia intensiva – *odds ratio* 2,17 (IC 95%: 1,01–4,68) – e expostos a procedimentos gerados de aerossóis, com *odds ratio* 1,36 (IC 95%: 1,21–1,52) (GÓMEZ-OCHOA *et al.*, 2021; NASA *et al.*, 2022; BANSAL *et al.*, 2025).

Gotículas são partículas respiratórias expelidas ao falar, tossir ou espirrar, com diâmetro $> 5 \mu\text{m}$, que tendem a se depositar rapidamente por gravidade e geralmente alcançam distâncias inferiores a 1–2 metros. Em contraste, aerossóis têm diâmetro $< 5 \mu\text{m}$, permanecem suspensos no ar por períodos prolongados, dispersam-se por distâncias maiores e apresentam maior probabilidade de serem inalados profundamente nas vias aéreas (SIEGEL *et al.*, 2007). Evidências recentes demonstram que a maior parte das partículas expelidas durante respiração, fala ou procedimentos médicos são aerossóis, e que a transmissão ocorre predominantemente a curta distância devido à alta concentração próxima à fonte emissora (FENNELLY, 2020).

Procedimentos geradores de aerossóis (PGA) são definidos como intervenções médicas que produzem partículas respiratórias suficientemente pequenas ($< 5 \mu\text{m}$) para permanecerem suspensas no ar por tempo prolongado, aumentando o risco de transmissão de patógenos respiratórios em ambientes hospitalares (SIEGEL *et al.*, 2007). Entre eles incluem-se intubação ou extubação, aspiração de vias aéreas, ventilação não invasiva, reanimação cardiopulmonar, broncoscopia e administração de medicamentos nebulizados (TRAN *et al.*, 2012). Além desses fatores, estudos recentes ressaltam que o risco de transmissão também é influenciado pelo contexto clínico, pela carga viral do paciente, pela proximidade física e pelas condições de ventilação do ambiente (NASA *et al.*, 2022; THURESSON *et al.*, 2022).

Mais especificamente em UTIP, a combinação entre a alta frequência de procedimentos geradores de aerossóis e o contato próximo com pacientes pediátricos — que frequentemente choram, tosem e têm menor capacidade de cooperar com barreiras físicas —

parece aumentar substancialmente o risco ocupacional quando comparado a outros setores hospitalares (HE *et al.*, 2017).

2.3.2 Estratégias de biossegurança

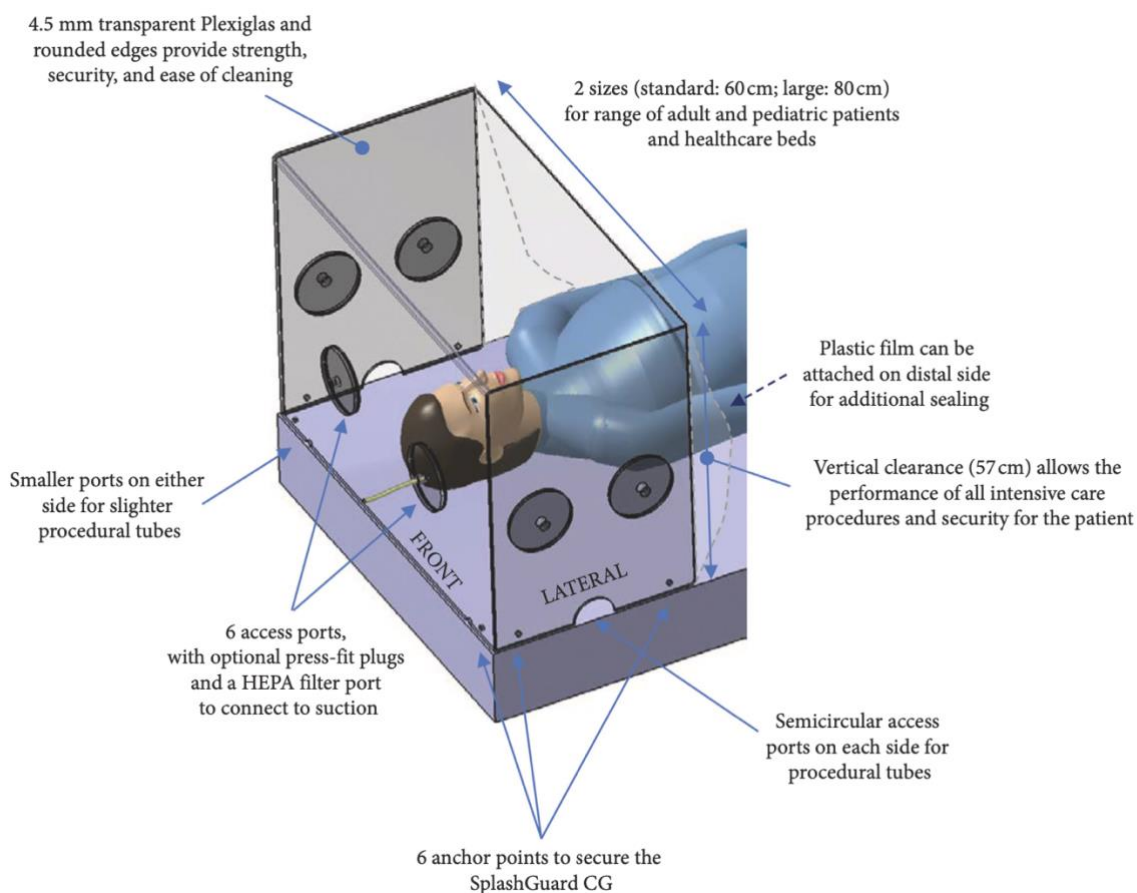
A implementação de práticas rigorosas de biossegurança mostrou-se determinante para a redução de infecções ocupacionais. Essas práticas incluem: (1) estratégias de proteção individual, (2) controle ambiental com ventilação adequada, (3) otimização da engenharia dos ventiladores, minimizando ao máximo vazamentos, e (4) organização de fluxos assistenciais (KLOMPAS *et al.*, 2021).

O uso adequado e consistente de equipamentos de proteção individual (EPI) envolve a utilização de máscaras N95 durante procedimentos geradores de aerossóis ou em períodos de alta incidência de vírus respiratórios, enquanto as cirúrgicas podem ser empregadas em situações de menor risco. Além disso, a proteção ocular, avental e luvas são recomendados para o contato com qualquer paciente portador de infecção respiratória (NASA *et al.*, 2022; ZANELLA; RHEE; KLOMPAS, 2025).

As diretrizes da *American Heart Association* (2025) reforçam o uso rigoroso de EPI — incluindo proteção respiratória (N95 ou equivalente), proteção ocular, luvas e avental — durante a realização de procedimentos geradores de aerossóis em pacientes com patógenos respiratórios de alto risco, como COVID-19 e influenza (CAO *et al.*, 2025). A vacinação universal de profissionais e pacientes elegíveis contra influenza e COVID-19 constitui estratégia comprovada para reduzir a transmissão e a gravidade das infecções (ZANELLA; RHEE; KLOMPAS, 2025).

A pandemia de COVID-19 impulsionou a adoção de novos protocolos de segurança, incluindo filtros de ar de alta eficiência (HEPA - *High-Efficiency Particulate Air*) nos circuitos respiratórios, modificações nas interfaces de ventilação, realização de procedimentos em salas com pressão negativa e o SGCG (JOUVET; AUBIN, [s. d.]; MONROE *et al.*, 2022; CAO *et al.*, 2025). Este último corresponde a uma caixa de acrílico colocada sobre a cabeça e o tronco do paciente com aberturas laterais que permitem o atendimento e que após permanecem fechadas e conectadas a pressão negativa (**Figura 1**) (FRANÇOIS *et al.*, 2020). Ainda, recomenda-se limitar o número de profissionais presentes durante os procedimentos e otimizar a ventilação ambiental, uma vez que a proximidade do paciente, a elevada carga viral e a ventilação inadequada são fatores críticos para a transmissão aérea (THURESSON *et al.*, 2022; UYEKI *et al.*, 2022).

Figura 1 - Características do protótipo final do *SplashGuard Caregiver*.



Fonte: (FRANÇOIS *et al.*, 2020)

2.4. Integração entre mecânica respiratória e biossegurança

A segurança do paciente em UTIP compreende diversas esferas de cuidado, com destaque para a ventilação pulmonar protetora, a prevenção de pneumonia associada à VM e o manejo criterioso das sedações, visando minimizar quadros de delirium e abstinência (KLOMPAS *et al.*, 2022; SMITH *et al.*, 2022; KNEYBER *et al.*, 2023). Essas estratégias são complementadas por medidas de prevenção de eventos adversos e de controle de infecção/biossegurança (SIEGEL *et al.*, D'ERRICO *et al.*, 2022). Um dos focos desta tese é a integração de dois pilares essenciais na segurança do paciente: a mecânica pulmonar e a biossegurança.

Através de uma análise sistemática da mecânica respiratória, torna-se possível identificar fenótipos fisiopatológicos distintos, como os padrões obstrutivos ou restritivos, que orientam a individualização dos ajustes ventilatórios. Esse manejo ocorre frequentemente em

contextos em que a realização de procedimentos geradores de aerossóis (PGA) é inevitável. Nesse cenário, a implementação de intervenções que reduzam a dispersão de partículas torna-se essencial, sendo o dispositivo SGCG uma estratégia possível visando um ambiente mais protegido, sem a necessidade de interromper a continuidade do cuidado do paciente crítico.

Essa articulação entre a fisiologia respiratória, a ventilação protetora e as práticas de biossegurança configuram um modelo ampliado de segurança na UTIP. Tal modelo reforça que a qualidade da assistência respiratória não se restringe apenas ao ajuste técnico de parâmetros no ventilador, mas depende da criação de um ambiente assistencial seguro que viabilize a execução dessas estratégias. Ao conectar esses conceitos, é possível avançar para um modelo de cuidado que seja, simultaneamente, eficiente, individualizado e sustentável para o paciente pediátrico crítico.

3 JUSTIFICATIVA

As disfunções respiratórias agudas constituem uma das principais causas de internação e mortalidade em unidade de terapia intensiva pediátrica. Apesar dos avanços no suporte ventilatório, o manejo desses pacientes ainda enfrenta importantes desafios relacionados à heterogeneidade da fisiopatologia pulmonar e à escassez de evidências específicas para a população infantil. A compreensão e o uso sistemático da mecânica respiratória têm se mostrado como uma das ferramentas fundamentais para a individualização da ventilação mecânica e para a redução de complicações associadas.

Paralelamente, a realidade assistencial em unidade de terapia intensiva pediátrica tem sido marcada por cenários de alta transmissibilidade viral, nos quais a segurança da equipe de saúde torna-se um componente indissociável da qualidade do cuidado. Situações como a pandemia de COVID-19 evidenciaram a vulnerabilidade dos profissionais expostos a aerossóis durante a assistência e a necessidade de equilibrar medidas de biossegurança com a continuidade do atendimento crítico.

Assim, esta tese se justifica pela necessidade de compreender, sistematizar e propor modelos que articulem o conhecimento científico da mecânica respiratória com práticas de biossegurança, contribuindo para a evolução das práticas assistenciais e para a formação de equipes mais conscientes e protegidas.

4 OBJETIVOS

4.1 OBJETIVO GERAL

Investigar e integrar os princípios da mecânica respiratória e da segurança ocupacional no manejo das disfunções respiratórias agudas em terapia intensiva pediátrica, propondo estratégias que aprimorem o cuidado ventilatório e reduzam os riscos ocupacionais em contextos de alta transmissibilidade viral.

4.2 OBJETIVOS ESPECÍFICOS

4.2.1 Caracterizar os parâmetros de mecânica respiratória em lactentes com disfunções respiratórias agudas graves, avaliando a relação entre pressão de pico, pressão de platô, complacência do sistema respiratório e resistência das vias aéreas durante a ventilação mecânica.

4.2.2 Identificar fatores fisiológicos que influenciam a discrepância entre pressão de pico e pressão platô, propondo indicadores objetivos para o ajuste individualizado da ventilação mecânica em pediatria.

4.2.3 Avaliar a confiabilidade de um modelo experimental *in vitro* na estimativa da pressão de platô a partir da pressão de pico em lactentes com disfunção respiratória aguda.

4.2.4 Avaliar a eficácia de um dispositivo de contenção de aerossóis, o *SplashGuard Caregiver*, na redução da dispersão de partículas e na exposição ocupacional de profissionais em unidade de terapia intensiva pediátrica.

4.2.5 Simular, por meio de modelagem computacional e dinâmica de fluidos, os padrões de movimentação de partículas em diferentes cenários ventilatórios e de arquitetura hospitalar.

4.2.6 Integrar os achados clínicos e de simulação em um modelo teórico-prático de manejo ventilatório seguro, conciliando proteção do paciente e da equipe multiprofissional.

5 MÉTODOS

Esta tese foi desenvolvida em duas etapas. O **primeiro artigo** baseou-se em dados provenientes da Unidade de Terapia Intensiva Pediátrica do Hospital de Clínicas de Porto Alegre, integrando a linha de pesquisa em mecânica ventilatória estabelecida por importantes estudos prévios (ANDREOLIO et al., 2021; BRUNO et al., 2022). A concepção desse artigo teve origem em um editorial que discutia a correlação entre a pressão de pico e a pressão platô em pacientes com doença restritiva (BURATTI; PIVA; JOUVET, 2022). O **segundo artigo** foi desenvolvido a partir de dados coletados no Hospital Sainte-Justine, em Montreal, como parte do *Fellowship* de Pesquisa em Fisiologia do Suporte Respiratório realizado na mesma instituição, vinculada à Universidade de Montreal de setembro de 2021 a junho de 2022 (ANEXO A).

5.1 MÉTODOS - ARTIGO 1

O estudo teve delineamento observacional retrospectivo e foi conduzido na Unidade de Terapia Intensiva Pediátrica do Hospital de Clínicas de Porto Alegre vinculado a Universidade Federal do Rio Grande do Sul, com aprovação do Comitê de Ética (CAAE 44889114.7.0000.5327). Foram incluídos dados de mecânica respiratória de lactentes de 1 a 24 meses internados entre maio de 2017 e dezembro de 2021 por falência respiratória aguda decorrente de infecção respiratória viral grave e submetidos à ventilação mecânica invasiva.

Os dados de mecânica respiratória utilizados nesta tese são provenientes dos bancos vinculados ao projeto de pesquisa intitulado “Valores da Mecânica Pulmonar de Crianças e Adolescentes Submetidos à Ventilação Mecânica”, aprovado pelo Comitê de Ética em Pesquisa do Hospital de Clínicas de Porto Alegre em julho de 2015 descrito acima. Em janeiro de 2022, foi submetida e posteriormente aprovada uma emenda (Emenda nº 1) com o objetivo de incluir novos dados de mecânica respiratória coletados rotineiramente durante o inverno de 2021.

A amostra incluiu pacientes com diagnóstico clínico e radiológico compatível com infecção respiratória viral grave, teste viral positivo e disponibilidade de ao menos uma mensuração completa de mecânica respiratória entre 6 e 48 horas após a intubação. Foram excluídos pacientes com doença pulmonar crônica, cardiopatias congênitas não corrigidas, aqueles com presença de drenos torácicos, traqueostomia ou fuga aérea superior a 15%.

A ventilação mecânica foi conduzida nos modos pressão controlada ou pressão regulada volume controlado, seguindo protocolo institucional previamente estabelecido, com

parâmetros de referência: frequência respiratória ≤ 20 incursões respiratórias por minuto, tempo inspiratório de 0,75–0,90 segundos, volume corrente de 8–10 mL/kg e PEEP de 3–5 cmH₂O. Para mensuração das variáveis respiratórias, assegurou-se ausência de esforço respiratório espontâneo e os pacientes foram temporariamente colocados em modo volume controlado mantendo os mesmos parâmetros, respeitando o exato volume que vinha sendo gerado a partir dos parâmetros de pressão controlada ou que estava pré-determinado no modo pressão regulada volume controlado. Foram realizadas pausas inspiratória e expiratória (de 0,5 - 5 segundos) para assegurar fluxo zero e permitir a obtenção de medidas estáticas P_{PLAT} , C_{ST} , resistência inspiratória (R_i) e resistência expiratória (R_e).

Nesta pesquisa avaliamos ainda a confiabilidade de um modelo experimental *in vitro*, proposto por Ilia et al. (2020), para estimar a P_{PLAT} a partir da PIP. A equação aplicada foi: $eP_{PLAT} = PIP - [5.067 - (0.858 \times C) - (0.018 \times R_i) - (0.390 \times (PIP - PEEP))]$. Os valores de eP_{PLAT} estimados foram comparados aos valores medidos *in vivo* (ILIA et al., 2020).

A análise estatística contemplou avaliação de normalidade (teste de Kolmogorov–Smirnov), teste t pareado para comparação entre PIP e P_{PLAT} , correlação de Pearson e análise de Bland-Altman para verificar concordância entre as variáveis medidas e estimadas. Adicionalmente, empregou-se regressão multivariada para propor uma equação de estimativa da P_{PLAT} a partir das variáveis de mecânica respiratória. Adotou-se nível de significância de $p \leq 0,05$, e todas as análises foram realizadas no software SPSS, versão 28.0 (IBM Corp., Armonk, NY, EUA).

5.2 MÉTODOS - ARTIGO 2

O estudo combinou uma coorte prospectiva com uma simulação computacional tridimensional, com objetivo de avaliar a eficácia do protótipo SGCG com vistas a redução do risco de transmissão aérea do COVID-19 em uma UTIP.

Etapa 1:

Coorte prospectiva: realizada entre abril e junho de 2021, na UTIP do Hospital Sainte-Justine, em Montreal, Canadá. Foram incluídos pacientes internados, inclusive adultos que acabaram utilizando leitos de UTIP em contexto de pandemia e os profissionais de saúde diretamente envolvidos na realização de PGA.

Os PGA analisados incluíram intubação e extubação orotraqueal, aspiração orofaríngea ou endotraqueal, inserção de sonda gástrica, uso de cânula nasal de alto fluxo e

ventilação não invasiva. Todos os profissionais utilizaram equipamentos de proteção individual (EPI) completos, conforme as recomendações da Organização Mundial da Saúde e dos Centros de Controle e Prevenção de Doenças, incluindo respirador N95, avental, luvas, proteção ocular e avental plástico adicional. Os quartos de UTIP foram divididos em dois grupos: (1) com o uso do dispositivo SGCG e (2) sem uso de SGCG. O SGCG consiste em uma caixa acrílica transparente posicionada sobre a cabeça e o tórax superior do paciente, contendo portas de acesso com filtros HEPA e sistema de sucção contínua, formando um microambiente de pressão negativa.

O desfecho primário foi a detecção de partículas virais de COVID-19 nas amostras ambientais e nos profissionais após a realização de AGP. As amostras foram coletadas de quatro pontos: (1) ar ambiente antes do AGP; (2) ar no interior do SGCG; (3) ar a um metro do paciente após AGP e (4) swab da testa de cada profissional ao término do procedimento. O coleta de material do ar se deu por cassetes com filtro de gelatina de 25 mm, operando a fluxos de 3 – 10 L/min, conforme o tipo de coleta, cujo armazenamento foi feito a -80° até a análise. A extração de RNA foi realizada utilizando o kit MagMAX Viral RNA Isolation (Applied Biosystems), e a quantificação do material genético viral foi conduzida por reação em cadeia da polimerase quantitativa (qPCR), direcionada ao gene ORF1b do COVID-19. Cada amostra foi analisada em triplicata, utilizando curva padrão plasmidial e limite de detecção de um plasmídeo por reação.

Em relação a análise estatística, os dados qualitativos foram expressos em frequências absolutas e percentuais, enquanto as variáveis quantitativas foram apresentadas como medianas, valores mínimos e máximos, sendo comparadas por meio do teste de Mann–Whitney U. As proporções entre os grupos (com e sem SGCG) foram avaliadas pelo teste exato de Fisher, considerando significância estatística para $p \leq 0,05$. Todas as análises foram realizadas no software SPSS, versão 21.0 (IBM Corp., Armonk, NY, EUA).

Etapa 2:

A segunda etapa consistiu na simulação do comportamento aerodinâmico das partículas em um ambiente virtual que reproduziu fielmente o quarto de UTIP utilizado nas coletas. Foi empregado o método de Dinâmica dos Fluidos Computacional (*Computational Fluid Dynamics* – CFD), desenvolvido por meio do software PowerFLOW, baseado na metodologia Lattice Boltzmann. O modelo tridimensional incluiu todas as características estruturais e de ventilação do ambiente (dimensões, entradas e saídas de ar, pressão negativa,

temperatura e taxa de troca de ar). Foram simulados três cenários: (1) paciente sem o uso do SGCG; (2) paciente com o SGCG apresentando uma abertura basal de 140 mm e (3) paciente com o SGCG apresentando uma abertura basal de 20 mm. As simulações consideraram um paciente adulto respirando a 12 L/min, com emissão de 100 partículas/L, distribuídas normalmente em torno de 3,2 μm de diâmetro. A trajetória e concentração das partículas foram analisadas ao longo do tempo, considerando a influência do fluxo de ar e a exposição potencial dos profissionais de saúde.

6 RESULTADOS (ARTIGOS)

Editorial: Peak Inspiratory Pressure to Estimate Plateau Pressure in Pressure Controlled Modes: Be Aware of Age and Disease.

Publicado no: Journal of Pediatric Critical Care Medicine em Março de 2022.

DOI: 10.1097/PCC.0000000000002896.

Número de citações: 5.

Driving pressure (ΔP = plateau pressure [P_{PLAT}] minus positive end-expiratory pressure [PEEP]) has generated increasing interest in recent years since it is considered the measure that best stratifies mortality risk in critically ill adults with acute respiratory distress syndrome (ARDS) (1). Consequently, ΔP is a potential predictor of ventilator-induced lung injury and a possible target to define the safety limits of mechanical ventilation (2, 3). In children, similar results have recently been published: Díaz et al (4) reported that ΔP is the formula that best discriminated patients with pediatric ARDS (PARDS) from those without, and van Schelven et al (5) showed that ΔP was independently associated with increased time to extubation.

P_{PLAT} measurement is performed in adults in the static condition when gas flow reaches zero using the volume-controlled mode (1). Contrasting with adult critical care, pressure-controlled modes with decelerating flow are predominantly used in the PICU (6), and peak inspiratory pressure (PIP) is used in some studies to estimate P_{PLAT} in PARDS for ΔP calculation (7–9). In this issue of Pediatric Critical Care Medicine, Patel et al (10) address a knowledge gap in the reliability of P_{PLAT} estimation using PIP in pressure-controlled mode. The authors show that in a cohort of 52 patients with PARDS (median age of 8.6 yr old [3.2–16.0 yr old]), PIP was close to P_{PLAT} measured in the static condition, that is, PIP was 1.0 ± 0.6 cm H₂O above P_{PLAT} (95% limits of agreement of -0.3 to 2.2). Therefore, Patel et al (10) concluded that PIP measured during decelerating flow ventilation may be an adequate surrogate of P_{PLAT} in PARDS when inspiratory flow approaches zero.

This new study by Patel et al (10) is well designed, including a cohort of PARDS children with an oxygenation index of 15.7 (9.0–21.0) and bilateral infiltrates on chest radiograph. In such a cohort, inspiratory airway resistance (R_{AW}) is most likely low, which is the ideal condition for a zero-flow plateau at the end of inspiratory time. Indeed, the inspiratory time constant (τ) in such circumstances is estimated to be 0.04 seconds. That is, since τ = respiratory system compliance $\times R_{AW}$, which is the same as this equation: (V_T [mL/kg] $\times$ weight

$[\text{kg}]/(\text{P}_{\text{PLAT}} - \text{PEEP}) (\text{cmH}_2\text{O}) \times \text{R} (\text{cmH}_2\text{O}/\text{L}/\text{s})$ (11). τ is arrived at by using median tidal volume = 6.4 mL/kg, mean body weight = 27 kg at 8.6 years old (12), $\text{P}_{\text{PLAT}} = 29.5 \text{ cmH}_2\text{O}$, $\text{PEEP} = 12 \text{ cmH}_2\text{O}$ (see article Supplemental Table in reference [10]), normal inspiratory resistance = 4 cmH₂O/L/s (13). As zero flow is obtained at five time constants ($\approx 0.2\text{s}$) (11), an inspiratory time set at more than 0.8 seconds (see the Supplemental Table in reference [10]) guarantees a prolonged period of time at zero flow and explains the quality of the current results.

Two main factors increase R_{AW} , and thus time constant, and lead to a lower correlation between PIP in the pressure-controlled mode and P_{plat} : age and respiratory diseases. Young age is associated with higher R_{AW} mainly because of narrower airways. For example, normal values of R_{AW} are 40 cmH₂O/L/s at birth, 15 cmH₂O/L/s at 1 year old, and 4 cmH₂O/L/s at 7 years old (i.e., 10-fold lower than the value at birth) (13). Hence, the longer time constant in younger children, who generally also have shorter inspiratory time. As one-third of children admitted to PICU are under 1 year old (14), the generalizability of the current results (10), which included older children, requires more investigation. Many respiratory diseases increase R_{AW} . When we do not have R_{AW} measurements available, it is still possible to qualitatively assess the presence of increased R_{AW} by using the respiratory disease classification of the Pediatric Mechanical Ventilation Consensus Conference, that is, normal lung, restrictive, obstructive airway, and mixed respiratory diseases (15).

Taking all of the above together, in the absence of P_{PLAT} measurement in the static condition, estimation of P_{PLAT} from dynamic measurements needs to also consider the resistive component. Much remains to be learned about the best ventilatory strategies in children, particularly with respect to protective ventilation: Do we need to target an optimal ΔP to improve outcomes? Does the age of the child alter the ΔP target? What is the impact of the respiratory disease on the ΔP goal? By offering a better understanding of the respiratory mechanics of critically ill children in different disease phenotypes, we should be able to answer these questions in the future.

References

1. Amato MBP, Meade MO, Slutsky AS, et al: Driving pressure and survival in the acute respiratory distress syndrome. *N Engl J Med* 2015; 372:747–755
2. Tonetti T, Vasques F, Rapetti F et al: Driving pressure and mechanical power: New targets for VILI prevention. *Ann Transl Med* 2017; 5:1–10

3. Saffaran S, Das A, Laffey JG, et al: Utility of driving pressure and mechanical power to guide protective ventilator settings in two cohorts of adult and pediatric patients with acute respiratory distress syndrome: A computational investigation. *Crit Care Med* 2020; 48:1001–1008
4. Díaz F, González-Damrauskas S, Cristiani F, et al: Driving pressure and normalized energy transmission calculations in mechanically ventilated children without lung disease and pediatric acute respiratory distress syndrome. *Pediatr Crit Care Med* 2021; 22:870–878
5. van Schelven P, Koopman AA, Burgerhof JGM, et al: Driving pressure is associated with outcome in pediatric acute respiratory failure. *Pediatr Crit Care Med* 2021 Oct 20.
6. Khemani RG, Smith L, Lopez-Fernandez YM, et al: Paediatric acute respiratory distress syndrome incidence and epidemiology (PARDIE): An international, observational study. *Lancet Respir Med* 2019; 7:115–128
7. Panico FF, Troster EJ, Oliveira CS, et al: Risk factors for mortality and outcomes in pediatric acute lung injury/acute respiratory distress syndrome. *Pediatr Crit Care Med* 2015; 16:e194–e200
8. Yehya N, Thomas NJ: Disassociating lung mechanics and oxygenation in pediatric acute respiratory distress syndrome. *Crit Care Med* 2017; 45:1232–1239
9. Rauf A, Sachdev A, Venkataraman ST, et al: Dynamic airway driving pressure and outcomes in children with acute hypoxemic respiratory failure. *Respir Care* 2021; 66:403–409
10. Patel B, Thomas NJ, Yehya N: Agreement Between Peak Inspiratory Pressure in Decelerating-Flow Ventilation and Plateau Pressure in Square-Flow Ventilation in Pediatric Acute Respiratory Distress Syndrome. *Pediatr Crit Care Med* 2022; 23:201–204
11. Alapont VM et al: Respiratory physiology. The physics of mechanical ventilation. In: *Handbook of Paediatric and Neonatal Mechanical Ventilation*. 2nd Edition, 2021, pp 65–117
12. WHO Multicentre Growth Reference Study Group: WHO Child Growth Standards: Methods and Development. Length/ Height-for_Age, Weight-for-Age, Weight-for-Length, Weight- for-Height and Body Mass Index-for Age: Methods and Development. 2007.
13. López-Herce Cid J: Respiratory function monitoring II: Compliance, airway resistance, dynamic hyperinflation, pulmonary dead-space, work of breathing. *An Pediatr* 2003; 59:278–285
14. Ibibebe I, Algert CS, Bowen JR, et al: Pediatric admissions that include intensive care: A population-based study. *BMC Health Serv Res* 2018; 18:1–8
15. Kneyber MCJ, de Luca D, Calderini E, et al: Recommendations for mechanical ventilation of critically ill children from the Paediatric Mechanical Ventilation Consensus Conference (PEMVECC). *Intensive Care Med* 2017; 43:1764–1780

Artigo 1: The lack of accuracy of peak inspiratory pressure to estimate plateau pressure in infants with severe obstructive lower airway disease.

Publicado no: Pediatric Pulmonology em agosto de 2024.

DOI: 10.1002/ppul.27245.

Número de citações: 3.

Abstract

Purpose: In children with acute respiratory distress syndrome receiving mechanical ventilation, the peak inspiratory pressure (PIP) is close to plateau pressure (P_{PLAT}) when inspiratory flow approaches zero. We aimed to evaluate the reliability of PIP to estimate P_{PLAT} in infants with severe respiratory viral infection (SRVI), characterized by increased airway resistance, and the accuracy of an equational model to estimates P_{PLAT} (eP_{PLAT}) based on PIP.

Methods: This was a retrospective observational study including mechanically ventilated children (1 to 24 month old) with SRVI, whose respiratory mechanics measurements were performed to evaluate PIP and P_{PLAT} . The measured P_{PLAT} was compared with the result of the equation: $eP_{PLAT} = PIP - [5.067 - (0.858 \times \text{static compliance}) - (0.018 \times \text{inspiratory resistance}) - (0.390 \times \text{pressure above positive-end expiratory pressure}) + (4.989 \times \text{inspiratory time})]$.

Results: Thirty-seven patients were included, with a median age of 3 (2–5) months. They presented a high inspiratory and expiratory resistance (136 ± 43 and 168 ± 66 cmH₂O/L/s, respectively) and a moderate reduction in static compliance: 0.75 ± 0.3 mL/kg/cmH₂O. PIP overestimated P_{PLAT} (33 ± 3 and 26 ± 5 cmH₂O, $p = 0.01$), with a mean difference of 7.3 ± 4 cmH₂O. Moreover, the Bland-Altman analysis demonstrated a mean difference between P_{PLAT} and eP_{PLAT} of 1.0 ± 4.0 cmH₂O, with 95% limits of agreement of -6.9 and 8.8 .

Conclusions: A large difference between PIP and P_{PLAT} was observed in infants with SRVI as well as the equation model was inaccurate to estimated P_{PLAT} based on PIP. Any estimation of P_{PLAT} from PIP needs to consider the resistive component of respiratory system.

Introduction

Plateau pressure (P_{PLAT}) is the pressure applied by a mechanical ventilator to the small airways and alveoli. In the absence of patient effort, P_{PLAT} reflects transpulmonary pressure and, depending on its values, may be associated with ventilation-induced lung injury (VILI). Thus, limiting P_{PLAT} represents a key principle in lung-protective ventilation in patients with acute respiratory distress syndrome (ARDS)¹. The 2023 Pediatric Acute Lung Injury Consensus Conference (PALICC-2)² proposed limiting P_{PLAT} , tidal volume (V_T), and driving pressure

(ΔP), the latter corresponding to the ratio between V_T and static compliance (C_{ST}), calculated at the bedside as P_{PLAT} minus positive-end expiratory pressure (PEEP).

Despite the lack of data on lung mechanics in pediatric patients, it is reasonable to infer that monitoring physiologic lung variables (e.g.: lung compliance and airway pressures) would be helpful to guide safer mechanical ventilation strategies. However, only recently data on pulmonary mechanics have been further explored, both in bench models and in different clinical populations, and predominantly in conditions of reduced compliance and/or increased resistance³⁻⁶. Therefore, different ventilation modes and mechanical ventilator models have been used, with the aim of understanding the behavior of lung mechanics in pediatric patients and how their use could be better applied in clinical practice.

Pulmonary compliance estimates the elasticity of the respiratory system, correlating the V_T to the pressure required to introduce it into the lungs. It can be measured statically (C_{ST}), which considers the elasticity of the lungs and chest compartment at rest (zero flow), using the formula $C_{ST} = V_T / (P_{PLAT} - \text{PEEP})$, or its dynamic measurement (C_D), which also considers airway resistance, using the formula $C_D = V_T / (\text{PIP} - \text{PEEP})$. Pulmonary resistance (R_{AW}) corresponds to the combined resistance from airways, lung tissue, and the endotracheal tube in invasive mechanical ventilation and is also influenced by the patient's spontaneous breathing. It can be calculated as $R = (\text{PIP} - P_{PLAT}) / \text{Flow (inspiratory or expiratory)}$ ^{7,8}.

Regarding P_{PLAT} , it is usually measured in volume control mode during an inspiratory pause, when the inspiratory flow approaches zero in a patient without respiratory effort⁹. In children receiving invasive mechanical ventilation (IMV), pressure-regulated volume control (PRVC) or pressure control (PC) are the most frequently used ventilation modes¹⁰, in which the flow and volume will vary according to the patient's respiratory mechanics and effort. Therefore, even with an inspiratory pause and absence of patient effort, peak inspiratory pressure (PIP) will be similar to P_{PLAT} only if the inspiratory time (T_I) is long enough to reach zero flow¹¹. However, in volume-controlled (VC) mode, the pressure varies according to respiratory mechanics, as the ventilator delivers a predefined V_T using the same flow-time waveform in every breath. Thus, during the inspiratory pause, the airway pressure will drop until it reaches P_{PLAT} , regardless of inspiratory time¹¹. Recent data from a group of pediatric patients with ARDS showed excellent agreement between P_{PLAT} and ΔP measurements, regardless of the ventilator mode used^{6,12}.

In general, P_{PLAT} is rarely measured in children, and PIP has been adopted as the main pressure parameter monitored¹³. Nonetheless, PIP is known to be significantly influenced by the Resistance properties of the respiratory system and may overestimate P_{PLAT} ³. The child's age and underlying respiratory disease are the main factors associated with increased R_{AW} and may lead to a poor correlation between PIP and P_{PLAT} . For instance, R_{AW} decreases with age as airway size increases (40 cmH₂O.L⁻¹.s at birth, 15 cmH₂O.L⁻¹.s at 1 year old [yo], and 4 cmH₂O.L⁻¹.s at 7 yo¹⁴), but increases in cases of airway narrowing such as in severe respiratory viral infection (SRVI)^{4,5}.

Given that one-third of children admitted to the pediatric intensive care unit (PICU) are under 2 yo and SRVI is the leading cause of admission in this age group¹⁵, we aimed to assess the reliability of PIP for estimating P_{PLAT} and whether a proposed in vitro model³ would be able to estimate P_{PLAT} from PIP in infants.

Methods

Study design and patient selection: This retrospective observational study was approved by the Institutional Review Board of Hospital de Clínicas de Porto Alegre (CAAE 44889114.7.0000.5327). Informed consent was waived due to the retrospective nature of the study, which involved routinely collected data. We included patients admitted to the PICU between May 2017 and December 2021 who met the following criteria: (1) acute respiratory failure requiring IMV; (2) diagnosis of SRVI based on classic symptoms of acute viral illness, chest radiograph showing typical signs of pulmonary hyperinflation associated or not with areas of consolidation, atelectasis and bronchoalveolar infiltrate, and a confirmed viral screening test; (3) age between 1 month and 2 years; and (4) at least one measurement of respiratory mechanics between 6 and 48 h after intubation. The following exclusion criteria were used: (1) chronic lung or airway disease; (2) uncorrected congenital heart disease; (3) chest or mediastinal drains considering possible air leaks; (4) tracheostomy; and (5) endotracheal tube air leak >15% [(inspiratory tidal volume minus expiratory tidal volume/ inspiratory tidal volume) × 100].

SRVI management and data collection: IMV support was administered using the Maquet® SERVO I ventilator, and the ventilatory modes used were PRVC and PC. The SRVI IMV management protocol, previously published elsewhere¹⁶, included the following parameters: low respiratory rate (RR à 20 breaths/minute), T_I between 0.75 and 0.90s, V_T between 8 and 10 mL/kg of ideal body weight, and positive end-expiratory pressure (PEEP)

between 3 and 5 cmH₂O. Demographics, laboratory and radiologic data, ventilator settings, and respiratory mechanics parameters were collected from medical records. After confirmation that the patient had no spontaneous respiratory effort (RR from the patient is equal to the RR setting in the ventilator), the ventilatory mode was switched to VC, with some exhaled V_T maintained and all other settings unchanged, for routine measurement of P_{PLAT}. If the patient exhibited spontaneous ventilation effort, the medical staff made individualized decisions regarding the best strategy to obtain measurements in static mode (without interference from voluntary respiratory efforts). In some cases, the ventilator's respiratory rate was temporarily increased (to elevate minute ventilation) and/or an additional dose of sedative was administered. Then, inspiratory and expiratory pauses were performed, both with a maximum duration of 5 s, and zero flow was confirmed by graphic evaluation in the ventilator. During the inspiratory pause, C_{ST}, inspiratory resistance (R_I), and P_{PLAT} were measured, while total PEEP and expiratory resistance (R_E) were recorded during the expiratory pause.

Assessment of the reliability of an in vitro model to estimate plateau pressure: Ilia et al.³ described the following equation based on a bench test model: estimated P_{PLAT} (eP_{PLAT}) = PIP – [5.067 - (0.858 × C) - (0.018 × R_{AW}) – (0.390 × pressure above PEEP) + (4.989 × T_I)]. This model was compared to the in vivo measurements obtained in our study, in which we used the absolute value of C_{ST}, i.e, not divided by weight, R_{AW} as R_I, and pressure above PEEP as PIP minus PEEP.

Statistical analysis: Data are expressed as absolute values, percentages, mean (with standard deviation), or median (with interquartile range). The Kolmogorov-Smirnov test was used to determine normality. The paired t-test was performed to evaluate the difference in the means of PIP and P_{PLAT}. Pearson correlation (r), coefficient of determination (r²), and Bland-Altman analyses were used to describe the agreement between PIP and P_{PLAT} and between P_{PLAT} and eP_{PLAT}. Moreover, a multivariate analysis was performed to estimate P_{PLAT} from PIP using our data. The level of significance was p-value ≤ 0.05. Spreadsheet, statistical analyses, and graphs were generated using SPSS, version 28.0.

Results

Thirty-seven subjects were included (**Table 1**) with a median age of 3.0 months (2.0 – 5.0), a Pediatric Index of Mortality 2 of 0.2% (0.2 – 0.5), and an oxygen saturation index of 4.0 (3.6 – 4.8). A causal virus was identified in 31 (83.8%) subjects, associated bacterial pneumonia

accounted for sixteen (43.2%) cases, and two (5.4%) patients developed pediatric acute respiratory distress syndrome (PARDS) according to the PALICC definition¹⁷.

Mean R_I and R_E were elevated (136 ± 43 and 168 ± 66 $\text{cmH}_2\text{O} \cdot \text{L}^{-1} \cdot \text{s}$, respectively) with a moderate decrease in C_{ST} (0.75 ± 0.3 $\text{cmH}_2\text{O} \cdot \text{L}^{-1} \cdot \text{s}$). The resulting median inspiratory time constant ($CT = C_{ST} \times R_I$) was 0.49 seconds ($0.42 - 0.72$) with a T_I of 0.80 ($0.80 - 0.85$) seconds and an RR of 18 ($18 - 20$) breaths/minute.

We observed that PIP (33 ± 3 cmH_2O) overestimated P_{PLAT} (26 ± 5 cmH_2O , $p < 0.001$) with a moderate correlation ($r = 0.57$, $p < 0.001$ and $r^2 = 0.33$, $p < 0.001$) and a mean difference between PIP and P_{PLAT} of 7.30 ± 4.0 cmH_2O (min., 1.0 and max., 14.0 cmH_2O) (**Figure 1, panel A**).

We found a moderate correlation ($r = 0.55$ and $r^2 = 0.30$) between eP_{PLAT} 25 ± 5 cmH_2O and P_{PLAT} 26 ± 5 cmH_2O . The Bland-Altman analysis demonstrated a mean difference between P_{PLAT} and eP_{PLAT} of 1.0 ± 4.0 cmH_2O with 95% limits of agreement (LOA) of -6.9 and 8.8 (**Figure 1, panel B**). In addition, a multivariate analysis was performed with the following variables from our sample: C_{ST} , R_I , T_I and pressure above PEEP. The purpose was to provide an equation that would calculate P_{PLAT} from PIP, but we achieved a maximum adjust r^2 of 0.34.

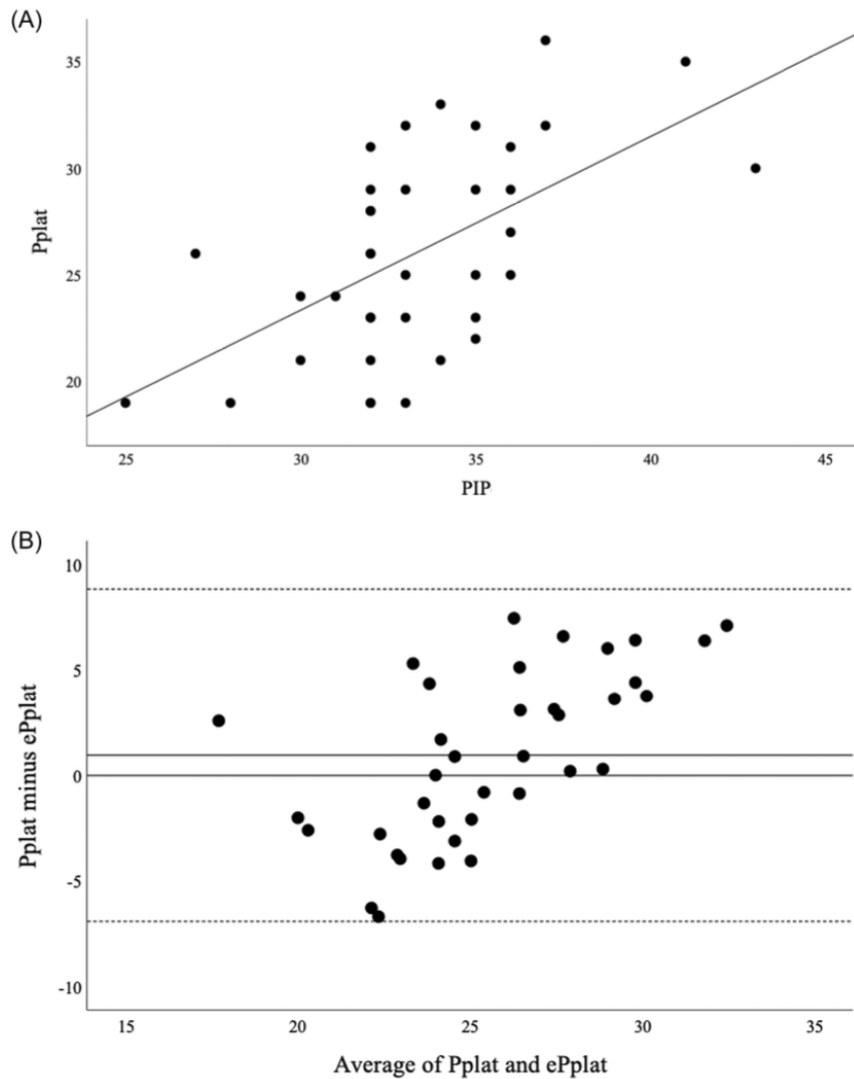
Table 1: Patients' characteristics.

Variable	N = 37
Age (months), median (IQR)	3.0 (2.0 – 5.0)
Weight (kg), median (IQR)	5.4 (4.5 – 7.0)
PIM-2 (% of mortality risk), median (IQR)	0.2 (0.2 – 0.5)
Male, n (%)	25 (67.6%)
Oxygen saturation index, median (IQR)	4.0 (3.6 – 4.8)
Tube size (cm), median (IQR)	3.5 (3.5 – 4.0)
Virus identified, n (%)	31 (83.8%)
One virus	22 (71.0%)
RSV, n	20
Parainfluenza, n	1
Influenza, n	1
Two viruses, n (%)	8 (25.8%)

RSV and adenovirus, n	1
RSV and parainfluenza, n	2
RSV and influenza, n	4
RSV and COVID-19, n	1
Three viruses, n (%)	1 (3.2%)
RSV and Influenza and Parainfluenza, n	1
Secondary diagnosis, n (%)	28 (75.7%)
Bacterial pneumonia, n (%)	16
Non-pulmonary sepsis, n (%)	12
PARDS, n (%)	2 (5.4%)
Comorbidities, n (%)	10 (27.0%)
Prematurity, n	7
Others, n*	3
Ventilation characteristics in decelerating-flow modes	
Ventilation mode	
PC-IMV, n (%)	23 (62.2%)
PRVC, n (%)	14 (37.8%)
Peak inspiratory pressure (cm/H ₂ O), median (IQR)	33 ± 3
Positive end-expiratory pressure (cm/H ₂ O), median (IQR)	5 (5 - 6)
Respiratory rate (breaths per minute), median (IQR)	18 (18-20)
Fraction of inspired oxygen (%), median (IQR)	0.30 (0.30 – 0.35)
Mean airway pressure (cm/H ₂ O), median (IQR)	12 (11 – 13)
Inspiratory time (seconds), median (IQR)	0.80 (0.80 – 0.85)
Tidal volume (ml/kg), median (IQR)	10 (10 – 11)
Saturation (%), median (IQR)	100 (96 – 100)
Ventilation characteristics at respiratory mechanics assessment in VC-CMV	
Peak inspiratory pressure (cm/H ₂ O), mean (SD)	36 ± 5
Plateau pressure (cm/H ₂ O), mean (SD)	26 ± 5
Static compliance (ml.kg ⁻¹ . cm/H ₂ O ⁻¹), mean (SD)	0.75 ± 0.3
Inspiratory resistance (cm/H ₂ O.L ⁻¹ .sec), mean (SD)	136 ± 43
Expiratory resistance (cm/H ₂ O.L ⁻¹ .sec), mean (SD)	168 ± 66
Auto-PEEP (cm/H ₂ O), median (IQR)	5 (3.0 – 7.0)

N: absolute value; IQR: 25–75 interquartile range; SD: standard deviation; PIM-2: *Pediatric Index of Mortality-2*; RSV: respiratory syncytial virus; PARDS: pediatric acute respiratory distress syndrome; Others*: genetic syndrome (1), congenital toxoplasmosis (1) and short bowel syndrome (1); PC-IMV: Pressure-control intermittent mandatory ventilation; PRVC: Pressure regulated volume control; VC-CMV: Volume-control continuous mandatory ventilation.

Figure 1: Reliability of peak inspiratory pressure in decelerating-flow mode (PIP) to estimate plateau pressure during square-flow mode (P_{PLAT}). **(A):** Scatterplot of P_{PLAT} and PIP in the 37 subjects studied ($r = 0.57$, $p < 0.001$). **(B):** Bland-Altman analysis comparing P_{PLAT} and eP_{PLAT} calculated using the Ilia et al. equation³, where the average difference between P_{PLAT} and eP_{PLAT} was 1.0 ± 4.0 mmHg with 95% limits of agreement of -6.9 and 8.8 .



Discussion

Based on objective measurements of respiratory mechanics in infants with SRVI, we found that the difference between PIP and P_{PLAT} can be as high as 14 cmH₂O, which is mainly due to the increased resistance in components of the respiratory system. Additionally, the equation proposed by Ilia et al.³ was not sufficiently accurate to estimate P_{PLAT} from PIP in this group of patients.

Patel et al.¹² reported that, in children with PARDS (mean age 8.6 years), PIP could be a reasonable substitute for P_{PLAT} , as long as the inspiratory flow reaches zero. R_{AW} was probably low in that cohort, which created favorable conditions for accurate P_{PLAT} estimation. The estimated inspiratory time constant in their study was 0.04s¹⁸ Considering that zero flow is achieved after five-time constants,¹⁹ a T_I higher than 0.80 s¹² would ensure adequate time at zero flow and explain the quality of the results. In contrast, in our study, the time constant was 10 times higher (0.49 s), and T_I was 0.80 s. Even with IMV parameters optimized for the underlying disease, zero flow cannot be guaranteed to achieve a strong agreement between P_{PLAT} and PIP.

Furthermore, we searched for a model to estimate P_{PLAT} from PIP. Using the equation proposed by Ilia et al.,³ we obtained a moderate correlation between P_{PLAT} and eP_{PLAT} , with a mean difference of 1.0 ± 4.0 mmHg and large LOA. Moreover, while the in vitro equation by Ilia et al.³ achieved an adjusted r^2 of 0.84, we only obtained a value of 0.34 using the same variables, which significantly limits external validation. This discrepancy may be related to the complexity of reproducing the variability of different lung mechanics scenarios in a bench model, as well as to the fact that we tested the eP_{PLAT} equations on a homogenous population with a small sample size.

The scientific evidence supports maintaining spontaneous ventilatory efforts during IMV, even in patients with severe lung diseases. However, from our perspective, it seems more reasonable to measure respiratory variables in static mode, as it is less subject to bias. More accurate data on respiratory physiology helps classify and quantify different types of lung diseases (e.g.: restrictive, obstructive or mixed) and select the safest mode of mechanical ventilation.

We are aware that this study has some limitations. Conducting the study in a single center may have introduced some inherent biases, such as: local characteristics (e.g.: virus epidemiology and/or genetic pattern of this sample) and severity of lung disease in this sample. Despite these considerations, we believe our results clearly highlight the importance of airway resistance when estimating P_{PLAT} in young children with SRVI.

Clinical research into PARDS has enhanced lung-protective ventilation strategies, including the monitoring of driving pressure.^{6,20} In patients with SRVI, the next step is to use P_{PLAT} and other pulmonary mechanics parameters to monitor disease progression, provide objective markers of disease reversibility, and inform the decision-making process for initiating weaning.

Conclusion

In this study, we demonstrated that in infants with SRVI, manifested by elevated inspiratory and expiratory respiratory resistance, there is a significant difference between PIP and P_{PLAT}. Additionally, the equation model based on PIP lacked accuracy to estimate P_{PLAT}. We recommend that, particularly in young children with SRVI, any attempt to estimate P_{PLAT} from PIP consider the resistance component of respiratory system.

References

1. Slutsky AS, Ranieri VM. Ventilator-induced lung injury. *N Engl J Med*. 2013;369:2126-2136.
2. Emeriaud G, López-Fernández YM, Iyer NP, et al. Executive summary of the second international guidelines for the diagnosis and management of pediatric acute respiratory distress syndrome (PALICC-2). *Pediatr Crit Care Med*. 2023;24:143-168.
3. Ilia S, van Schelven PD, Koopman AA, et al. Effect of endotracheal tube size, respiratory system mechanics, and ventilator settings on driving pressure. *Pediatr Crit Care Med*. 2020;21:e47-e51.
4. Andreolio C, Piva JP, Bruno F, daRocha TS, Garcia PC. Airway resistance and respiratory compliance in children with acute viral bronchiolitis requiring mechanical ventilation support. *Indian J Crit Care Med*. 2021;25:88-93.
5. Bruno F, Andreolio C, Garcia PCR, Piva J. The relevance of airway resistance in children requiring mechanical ventilatory support. *Pediatr Crit Care Med*. 2022;23(10):e483-e488.
6. Cruces P, Moreno D, Reveco S, Ramirez Y, Díaz F. Plateau pressure and driving pressure in volume and pressure-controlled ventilation: comparison of frictional and viscoelastic resistive components in pediatric acute respiratory distress syndrome. *Pediatr Crit Care Med*. 2023;24(9):750-759.
7. Newth C, Khemani R. Respiratory monitoring of the mechanical behavior of the respiratory system during mechanical ventilation: clinical application. In: *Rimensberger PC Pediatric and neonatal mechanical ventilation*. Springer-Verlag; 2015:355-371.
8. Guerra SD, Morais CL. Métodos gráficos de monitorização da ventilação mecânica. In: *Piva JP, Garcia PCR. Medicina Intensiva em Pediatria*. Revinter; 2014:691-724.
9. Hess DR. Respiratory mechanics in mechanically ventilated patients. *Respir Care*. 2014;59:1773-1794.

10. Farias JA, Fernández A, Monteverde E, et al. Mechanical ventilation in pediatric intensive care units during the season for acute lower respiratory infection: a multicenter study. *Pediatr Crit Care Med*. 2012;13:158-164.
11. Rittayamai N, Katsios CM, Beloncle F, Friedrich JO, Mancebo J, Brochard L. Pressure-controlled vs volume-controlled ventilation in acute respiratory failure. *Chest*. 2015;148:340-355.
12. Patel B, Thomas NJ, Yehya N. Agreement between peak inspiratory pressure in decelerating-flow ventilation and plateau pressure in square-flow ventilation in pediatric acute respiratory distress syndrome. *Pediatr Crit Care Med*. 2022;23:201-204.
13. Bhalla AK, Klein MJ, Emeriaud G, et al. Adherence to Lung-Protective ventilation principles in pediatric acute respiratory distress syndrome: a pediatric acute respiratory distress syndrome incidence and epidemiology study. *Crit Care Med*. 2021;49: 1779-1789.
14. López-Herce Cid J. Monitorización de la función respiratoria en el niño con ventilación mecánica (II): complianza, resistencia, hiper- insuflación dinámica, espacio muerto y trabajo respiratorio. *Anales de Pediatría*. 2003;59:278-285.
15. Ibibebe I, Algert CS, Bowen JR, Roberts CL. Pediatric admissions that include intensive care: a population-based study. *BMC Health Serv Res*. 2018;18:264.
16. Ferlini R, Pinheiro FO, Andreolio C, Carvalho PRA, Piva JP. Characteristics and progression of children with acute viral bronchiolitis subjected to mechanical ventilation. *Rev Bras Ter Intensiva*. 2016;28:55-61.
17. Pediatric Acute Lung Injury Consensus Conference Group. Pediatric acute respiratory distress syndrome: consensus recommendations from the Pediatric Acute Lung Injury Consensus Conference. *Pediatr Crit Care Med*. 2015;16:428-439.
18. Buratti CR, Piva J, Jouvét P. Peak inspiratory pressure to estimate plateau pressure in pressure controlled modes: be aware of age and disease. *Pediatric Critical Care Medicine*. 2022;23:225-226.
19. Alapont VM, Villanueva AM, Carrascosa MA, Burren JM, Nair M, Calderini E. Respiratory physiology. The physics of mechanical ventilation In: Villanueva AM, Orive JP, Cuscó MG, Blokpoel R, Ódena MP, Rimensberger P *Handbook of Paediatric and Neonatal Mechanical Ventilation*. Tesela Ediciones; 2021:65-117.
20. Fernández-Sarmiento J, Bejarano-Quintero AM, Tibaduiza JD, et al. Time course of mechanical ventilation driving pressure levels in pediatric acute respiratory distress syndrome: outcomes in a prospective, multicenter cohort study from Colombia, 2018–2022. *Pediatr Crit Care Med*. 2024.

Artigo 2: Effectiveness of SplashGuard Caregiver prototype in reducing the risk of aerosol transmission in intensive care unit rooms of SARS-CoV-2 patients: a prospective and simulation study

Publicado em: Journal of Hospital Infection em novembro de 2023

DOI: 10.1016/j.jhin.2023.11.007.

Número de citações: 6.

Summary

Background: The contagiousness of severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) is known to be linked to the emission of bioaerosols. Thus, aerosol-generating procedures (AGPs) could increase the risk of infection among healthcare workers (HCWs).

Aim: To investigate the impact of an aerosol protection box, the SplashGuard Caregiver (SGCG) with suction system, by direct analysis of the presence of viral particles after an AGP, and by using the computational fluid dynamics (CFD) simulation method.

Methods: This prospective observational study investigated HCWs caring for patients with SARS-CoV-2 admitted to an intensive care unit (ICU). Rooms were categorized as: SGCG present and SGCG absent. Virus detection was performed through direct analysis and using a CFD model to simulate the movement dynamics of airborne particles produced by a patient's respiratory activities.

Findings: Of the 67 analyses performed, three samples tested positive on quantitative polymerase chain reaction: one of 33 analyses in the SCCG group (3%) and two of 34 analyses in the non-SGCG group (5.9%). CFD simulations showed that: (1) reduction of the gaps of an SGCG could decrease the number of emitted particles remaining airborne within the room by up to 70%; and (2) positioning HCWs facing the opposite direction to the main air flow would reduce their exposure.

Conclusions: This study documented the presence of SARS-CoV-2 among HCWs in a negative pressure ICU room of an infected patient with or without the use of an SGCG. The simulation will help to improve the design of the SGCG and the positioning of HCWs in the room.

Introduction

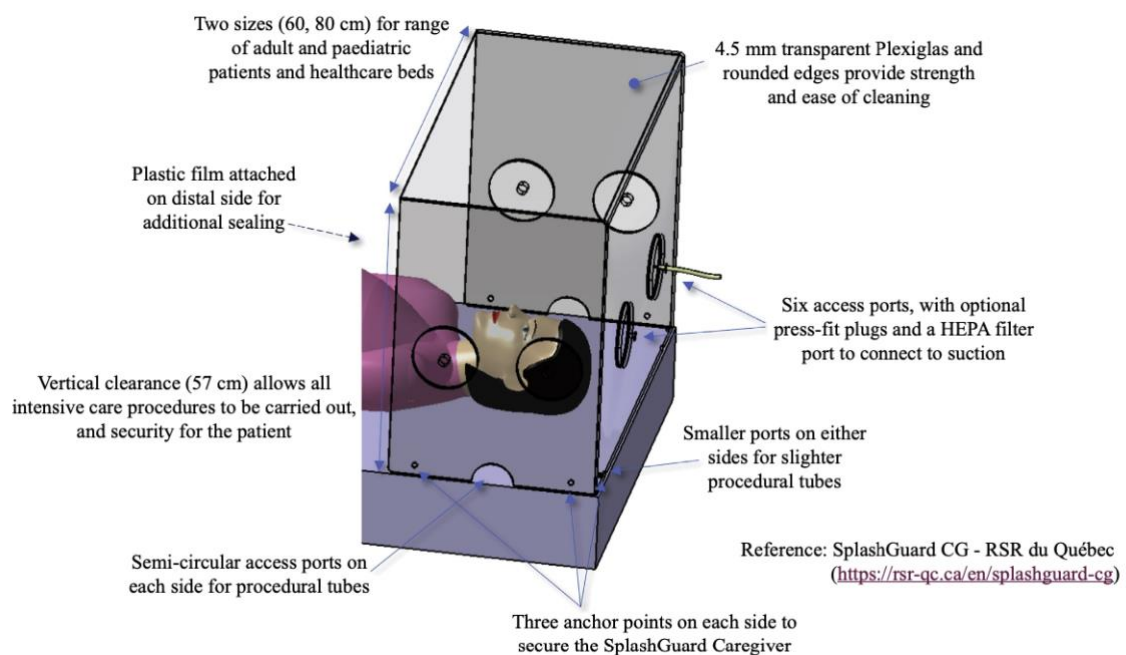
The coronavirus disease 2019 (COVID-19) pandemic, caused by severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2), has affected more than 760 million people worldwide, with 6.9 million deaths as of 1st June 2023 [1]. The contagiousness of SARS-CoV-2 is associated with the emission of viral aerosols by infectious patients, and by the fact that they can remain viable and infectious for hours and even days on surfaces [2,3]. The risk of healthcare workers (HCWs) developing infectious diseases because of contact with patients is well recognized [4] and increased during the COVID-19 pandemic [4-6].

Aerosol-generating procedures (AGPs) may increase the exposure of HCWs to airborne infectious agents [7,8], although there is a lack of evidence on the subject [9]. AGPs, along with normal respiratory activities such as breathing and coughing, lead to the shedding of airborne particles into the environment, and these can be virus-laden when emitted by an infectious patient [10]. The diameter of airborne particles ranges between 100 nm and 1 mm (droplets) before evaporating and forming a droplet nuclei or aerosol [11,12]. It is considered that particles with diameter <10 µm can remain airborne and initiate infection at both short and long distances and can propagate further than 1 m from the source, remaining infectious in the air for up to 3h [2,10]. High-risk AGPs are considered to be those that involve the manipulation of high viral load tissues, such as endotracheal intubation or extubation, airway suctioning, gastric tube insertion and non-invasive ventilation [13, 15]. However, a recent study showed that upper airway suctioning was not associated with a higher aerosol concentration, and this was actually lower compared with breathing and coughing [16].

As a result, clinicians have developed protective devices, such as the *SplashGuard Caregiver* (SGCG), with the aim of minimizing the exposure of HCWs to aerosols during AGPs, thus providing more freedom to perform these AGPs. The SGCG was proposed as a redesigned ‘aerosol box’, originally invented by a Taiwanese anaesthesiologist (**Figure 1**) [17-19]. It consists of a large Plexiglas box installed above the patient’s head and torso that contains six doors for simultaneous access by HCWs [18,19]. These openings have optional press-fit plugs and a high-efficiency particulate air (HEPA) filter, allowing them to be connected to continuous suction to create a negative environment, which provides an extra shield to minimize the spread of viral particles. Some reports described that the SGCG can reduce the exposure of HCWs to virus during endotracheal intubation in the operating theatre, and suctioning in the emergency room [15,20,21].

The aim of this study was to investigate the impact of the SGCG prototype on the presence of viral particles on HCWs' foreheads, and in the air near HCWs responsible for treating patients hospitalized with COVID-19 in an intensive care unit (ICU). Furthermore, it was proposed that the movement of aerosols produced by a patient in an ICU room should be modelled using three-dimensional (3D) computational fluid dynamics (CFD) [20] to simulate air flow as well as the movement of numerous airborne particles produced by a patient's respiratory activity (conventional breathing and coughing), with and without the use of the SGCG. From this, the impact of each scenario on the local concentration of potentially virus-laden airborne particles can be better understood, as it is essential to find ways to mitigate risks for frontline workers.

Figure 1. *SplashGuard Caregiver* prototype. Source: adapted from Jouvét and Aubin [19]. HEPA, high-efficiency particulate air.



Methods

A prospective single-center cohort study was undertaken from April to June 2021. The study included patients admitted to an ICU, who tested positive for COVID-19, and HCWs. The primary outcome was evidence of a decrease in the proportion of active SARS-CoV-2 viral particles in samples collected from HCWs' foreheads and in the patient's environment among those who used the SGCG during AGPs. Regardless of whether or not an SGCG was used, all HCWs wore the personal protective equipment (PPE) recommended by the World Health Organization and the Centers for Disease Control and Prevention during an AGP, which included N95 respirator, gown, gloves, eye protection (goggles or face shield) and apron. AGPs include endotracheal intubation/extubation, oropharyngeal or endotracheal aspiration, gastric tube placement, use of nasal cannula in spontaneous breathing, ventilatory assistance with high-flow nasal cannula, and other non-invasive ventilation [13]. The study was approved by the Ethics Committee of Centre Hospitalier Universitaire Sainte-Justine (No. MP-21- 2020-2870), and all patients provided written informed consent prior to inclusion in the study.

In a second step, a 3D CFD model [22] was made of an ICU room containing a virtual patient with and without an SGCG during normal breathing or coughing. This considered all morphological characteristics of the ICU room in terms of ventilation (negative pressure, temperature, inlet/outlet positions and mass flow rates) and architecture (exact room dimensions and geometry, door position, patient position and additional unit particularities). The 3D CFD model is able to evaluate the local concentration of particles present in the environment over time, considering their size (droplets and aerosols), initial position and spreading velocity (normal breathing or coughing), and environmental factors such as the air flow induced by the room's ventilation system. A commercial Lattice Boltzmann-based method code, PowerFLOW [23], was used to run the numerical simulations and generate numerical results. The underlying principles of this technique have been described in detail by Crawford et al. [22].

Protective measure against aerosols of viral particles (Figure 1): The SGCG is an acrylic box, placed over the patient's head and upper torso. It contains six different access doors (two at the front and two on each side) which allows several HCWs to perform procedures related to the upper airways simultaneously, while limiting the amount and distance of infected particles projected into the environment. In addition, two smaller semicircular openings on the side allow the introduction of the breathing circuit and suction equipment, as well as the oxygen tubing for bag-valve-mask ventilation or the non-rebreather mask. To provide extra protection against

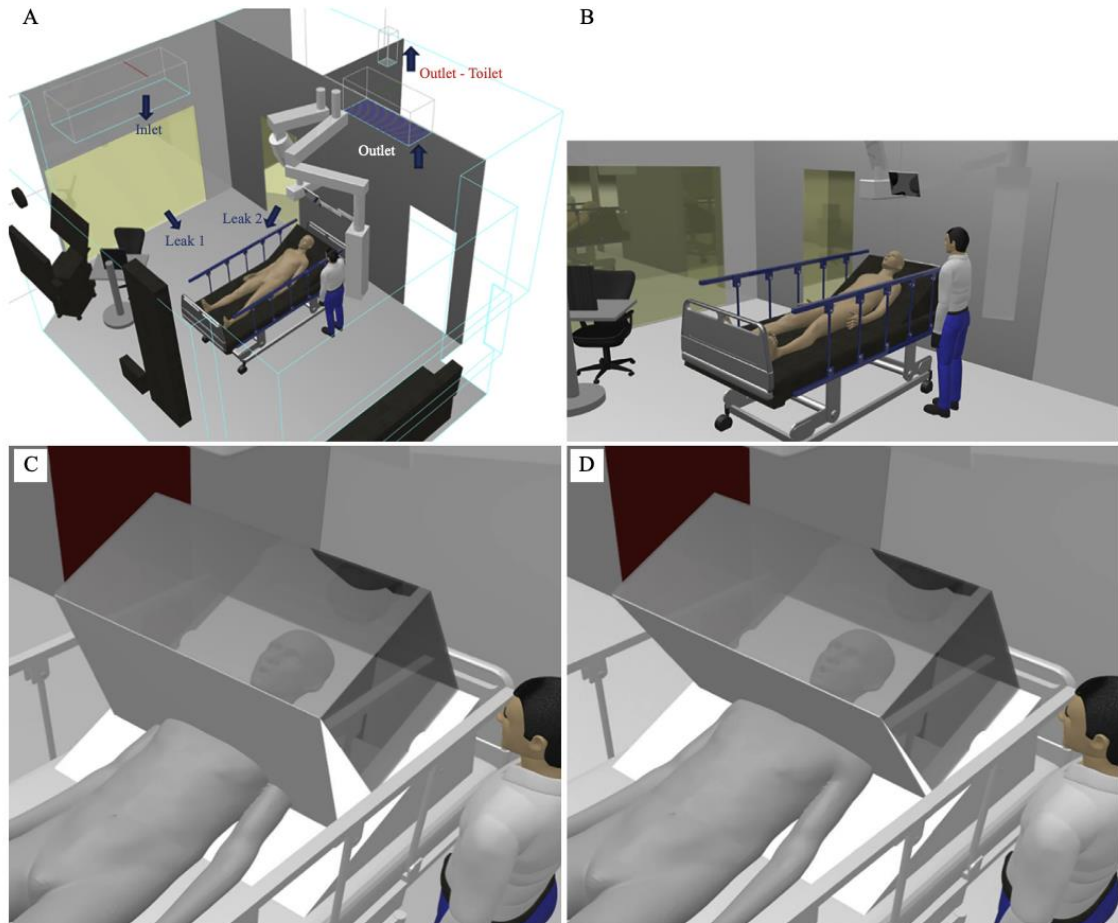
aerosolization, in the absence of procedures, shutters occlude the orifices, and a plastic film covers the rest of the patient's torso/pelvis for additional sealing. The shutters have optional press-fit plugs with a HEPA filter that can be connected to a negative pressure suction system, leading the air from inside the SGCG to the usual medical gas exhaust duct. Finally, the SGCG has six anchor points on the base, which provides more stability. The user manual detailing the use of the SGCG (Class 1 reusable medical device) can be consulted using the following link: <https://rsr-qc.ca/Splashguard-cg/>.

Virus detection: Environmental sampling for SARS-CoV-2 was performed using personal pumps and sampling cassettes: (1) in the ambient air before the AGP; (2) in the air of patients inside the SGCG; (3) in the air 1 m from the patient after the AGP; and (4) through a swab on the forehead of each HCW after an AGP. The techniques performed for viral detection are detailed below.

- Air samples: Stationary and personal air samples were collected using Institute of Occupational Medicine (IOM) air samplers loaded with 25-mm gelatin filters (SKC, Eighty Four, PA, USA). For stationary samples, a flow rate of 10 L/min was achieved using a calibrated regulator connected to suction present in the room [24]. Personal samples were collected at a flow rate of 3 L/min using a GilAir Plus Personal Air Sampling Pump (Sensidyne, Saint Petersburg, FL, USA). After collection, gelatin filters were dismantled from the IOM devices, solubilized in 2 mL of Gibco's viral transport medium (VTM; ThermoFisher, Winnipeg, Canada), and stored at -80°C until analysis.
- Swab samples The forehead of each HCW was swabbed using a HydraFlock 600 sterile flock swab (Puritan, Guilford, ME, USA) at the end of treatment. Swab samples were eluted in 1 mL of VTM and stored at -80°C until analysis.
- RNA extraction of samples: RNA was purified using 400 µL of each sample using a MagTMMAX Viral RNA Isolation Kit (Applied Biosystems, Vilnius, Lithuania). Purified RNA was eluted in 50 µL of elution buffer and stored at -80°C until quantification. A no template control was performed for each lot of samples threatened.
- Quantitative PCR: Briefly, each sample was amplified in triplicate, targeting ORF1b of the SARS-CoV-2 genome. Viral genome quantities were estimated by averaging the results of all replicates based on a ORF1b plasmid standard curve with a lower limit of detection of 1 plasmid per quantitative polymerase chain reaction (qPCR), corresponding to a cycle threshold value of 38. The protocol has been described in detail elsewhere [24,25].

3D computational fluid dynamics model [22]: The simulations were based on the digital twin of the same patient room in which the samples were taken, as described above. It includes the room furniture and devices in place, including the bed, as well as the position of the HCW. The ventilation details were also reproduced based on the technical specifications (**Figure 2A**), with an air mass flow outlet at the center of the room, above the bed, which extracts 1410 m³/h from the room. A secondary outlet is located in the bathroom and extracts 168 m³/h. These airflow rates were measured directly at each ventilation vent by the facilities department using an anemometer. In addition, an inlet is located above the main door and allows 935 m³/h of air inside the room. Finally, a 20-mm gap was modelled between both room doors and the floor in order to allow air to enter the room from the corridor, and balance the mass flows injected and extracted from the room. Overall, this corresponded to 11 air changes per hour within this ICU room. The baseline scenario corresponded to the patient lying on the bed without an SGCG (**Figure 2B**). Two modified scenarios were also modelled, corresponding to the same patient with an SGCG: one with a gap between the plastic film and the Plexiglas box of 140 mm at the base (**Figure 2C**); and one with a 20-mm gap (**Figure 2D**). The SGCG was modelled in the configuration where the negative pressure suction system is active. The patient was modelled in detail, considering skin temperature as well as a detailed geometry of the upper respiratory airways of an adult. In all three cases, the distribution of particles dispersed in the environment by the air flow was analyzed after being shed by the patient's normal breathing respiratory activities. The breathing rate was set at 12 L/min and the emission rate was chosen in order to have a concentration of emitted particles of 100 particles/L [26]. The size distribution used in the model for these emitted airborne particles, which are considered to be reduced to droplet nuclei, is a normal distribution centered around 3.2 μ m [27]. It was considered that each of these particles can contain one or more virions and is potentially infectious when inhaled in sufficiently large quantities. It should also be noted that the use of different oxygen support devices or the practice of a specific type of AGP were not considered in the simulation, and that only a standardized adult patient was modelled (no child patient was modelled).

Figure 2. Intensive care room simulation. (A) Room model. (B) Individual without *SplashGuard Caregiver* (SGCG). (C) First configuration of individual with SGCG (140-mm gap). (D) Second configuration of individual with SGCG (20-mm gap).



Statistical analysis: Qualitative data are expressed as absolute frequencies (n) and percentages. Continuous quantitative variables are expressed as medians (minimum and maximum) considering non-parametric distribution, and analysis was performed using the Mann Whitney U-test. The qPCR results of patients that used an SGCG and their HCWs were compared with samples taken from rooms of patients without an SGCG using Fisher's Chi-squared test. Statistical analyses were performed using SPSS Version 21.0 (IBM Corp., Armonk, NY, USA).

Results

Phase 1 (qPCR analysis): Fourteen batches of samples were included, and 67 analyses were performed in the room with SARS-CoV-2-positive patients. Median age was 11 years (range 1 month - 40 years) and median weight was 54 kg (range 3.5 - 107 kg). Only three samples were identified as qPCR positive: one in the SGCG group (one of 33 analyses: 3%) and two in the non-SGCG group (two of 34 analyses: 5.9%). The exposure time of the qPCR collection devices (pump or swab) did not differ significantly between the groups. On Chi-squared analysis, no difference was found between the groups. In addition, none of the HCWs were infected. Although feedback was not collected from patients and HCWs about their tolerance to the SGCS, there were no records of the need for early removal of the device when it was prescribed. **Table I** shows the details for individuals in the SGCG and non-SGCG groups, including age, weight, AGP perform

Table I: Characteristics of the sample and comparison in prevalence of quantitative polymerase chain reaction (qPCR) samples between *SplashGuard Caregiver* (SGCG) and non-SGCG groups

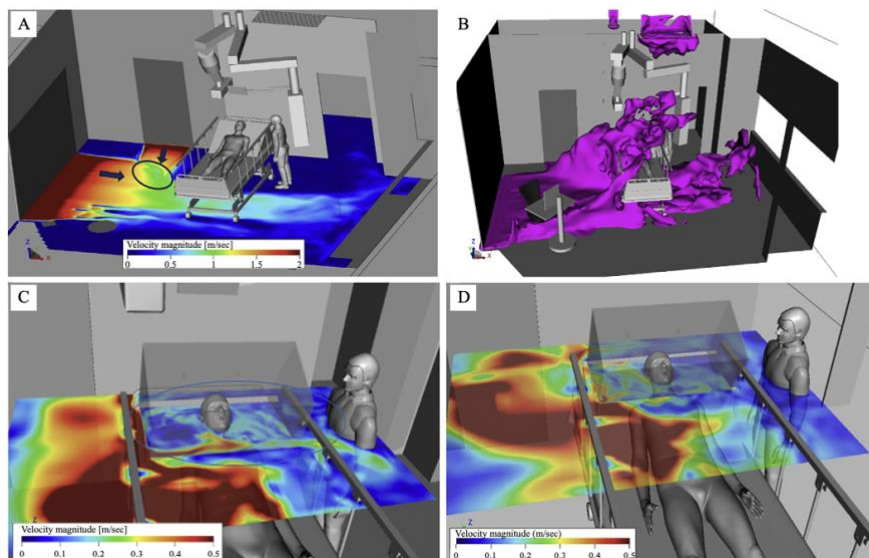
	SGCG absent (N=9)	SGCG present (N=5)	P-value
Age (years)	2 (1 month–40 years)	33 (21–40)	0.01 ^a
Weight (kg)	11 (3.5–70)	101 (80–107)	0.001 ^a
Procedure (N)			NA
	Transfer (N=1)	NIV (N=3)	
	HFNC (N=2)	HFNC (N=2)	
	Trach (N=2)		
	EITc (N=1)		
	HFOV (N=3)		
Before AGP			
Time exposure (min)	250 (203–290)	250 (180–320)	0.83 ^a
qPCR in room air (N)	Negative (N=9)	Negative (N=5)	NA
After AGP			
Time exposure (min)		240 (180–320)	NA
qPCR in SGCG air (N)		Negative (N=4)	NA
		NA (N=1)	
Time exposure (min)	35 (21–135)	60 (20–78)	0.52 ^a
qPCR in HCW air (N)	Negative (N=12)	Negative (N=11)	1.00 ^b
	Positive (N=1)	Positive (N=1)	
Time exposure (min)	2 (1–2)	2 (2–2)	0.69 ^a
qPCR in HCW forehead (N)	Negative (N=12)	Negative (N=12)	1.00 ^b
	Positive (N=1)	Positive (N=0)	

Time exposure, exposure time of the pump or swab of the respective qPCR collection; Transfer, transferring the patient to the ward; HFNC, care of patient with high flow nasal cannula; Trach, care of a patient with tracheostomy tube; EITc, endotracheal intubation care; HFOV, care of patient with high-frequency oscillatory ventilation; NIV, care of patient with non-invasive ventilation; AGP, aerosol-generating procedures; NA, not assessed; HCW, healthcare worker. P 0.050 was considered to indicate significance: (a) Mann-Whitney U-test; (b) Fisher's Chi-squared test. Data are presented as median (minimum and maximum), and number of subjects is presented as absolute value (N).

Phase 2 (simulation analysis): An aerodynamic analysis was conducted using CFD. Airflow leakages were observed through the thin gap below the two room doors, due to the room being under negative pressure (**Figure 3A**). Due to this gap and the position of the extraction vent of the heating, ventilation and air conditioning (HVAC) system above the patient, air flows enter the room under the two perpendicular room doors, colliding at high velocities (>1 m/s) before merging and propagating towards the ceiling on the opposite side of the room.

The SGCG and the patient are in the path of the turbulent flow which forms between the room doors and the HVAC outlet and deflects towards the ceiling (**Figure 3B**). It is important to note that this air flow is able to enter the SGCG through the opening created by the plastic film, creating strong recirculation inside, and can exit through the opposite opening, near the HCW (**Figure 3C,D**). The air renewal induced by the suction at the back of the device is not as efficient as that induced by the air flow passing through the opening created by the plastic film, regardless of the size of the gap (**Figure 3C,D**).

Figure 3. Simulation analysis. (A) Airflow leakage through the gap between the doors and the floor (two-dimensional map of velocity magnitude). (B) Airflow leakage inside the room and around the SplashGuard Caregiver (SGCG) [three-dimensional (3D) envelope of velocity magnitude >0.4 m/s]. (C) 3D visualization of the air flow entering the SGCG through the opening created by the plastic film (140-mm-gap scenario). (D) Air flow does not enter the SGCG through the reduced opening created by the plastic film (20-mm-gap scenario).



After performing the aerodynamic simulation and analysis, the particles emitted during the breathing and coughing cycle were introduced in the simulation and followed. In the first scenario (no SGCG), the potentially infectious exhaled particles follow the air flow described in **Figure 3C** and disperse throughout the room because of the mixing ventilation (**Figure 4A**). Over time, these particles are either extracted by the ventilation outlet, deposit on surfaces (walls, tables, ceiling, medical equipment) or remain airborne. In the second scenario (SGCG with a large gap), the emitted particles are trapped and accumulate within the protective device. However, due to the air flow going through the SGCG, as depicted in **Figure 4B**, these particles end up exiting through the opening on the right side of the SGCG, near the HCW. Furthermore, as the presence of the SGCG reduces the velocity of the air flow that reaches the other side (<0.4 m/s), the particles entrained stay near the HCW for a longer period (**Figure 4B**). Finally, in the third scenario (SGCG with a small gap), the particles tend to remain within the protective box and only a few escape through the gap. A quantitative analysis is presented in **Figure 5**, which represents the cumulative number of particles that go through the orange box modelled virtually around the HCW (**Figure 4D**), located on the left side of the patient. As shown in **Figure 1**, the number of particles passing within breathing distance of the HCW was reduced by 17% when an SGCG (with a gap between the plastic film and the Plexiglas box of 140 mm) was added compared with the baseline scenario, and reduced by 93% when the gap was reduced to 20 mm. Thus, placing a HCW behind the patient or to the right would lower their exposure to higher particle concentrations.

Figure 4. Three-dimensional visualization of the numerically modelled airborne particles emitted by a patient and dispersed throughout the room. (A) In the baseline configuration at t 25 s. (B) In the first *SplashGuard Caregiver* (SGCG) configuration (140-mm gap) at t 25 s. (C) In the second SGCG configuration (20-mm gap) at t 25 s. (D) Virtual representation of the breathing zone (orange box) around the healthcare worker.

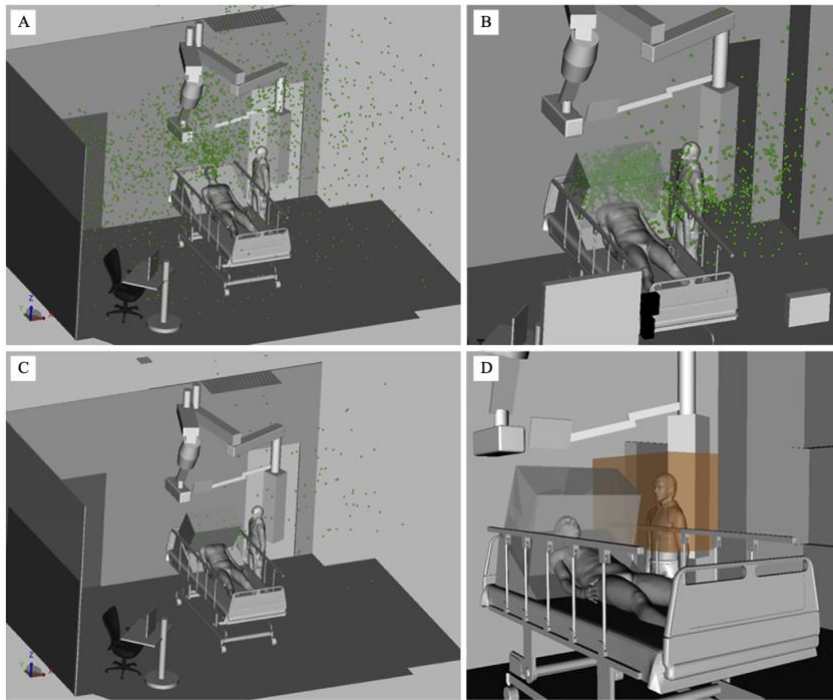
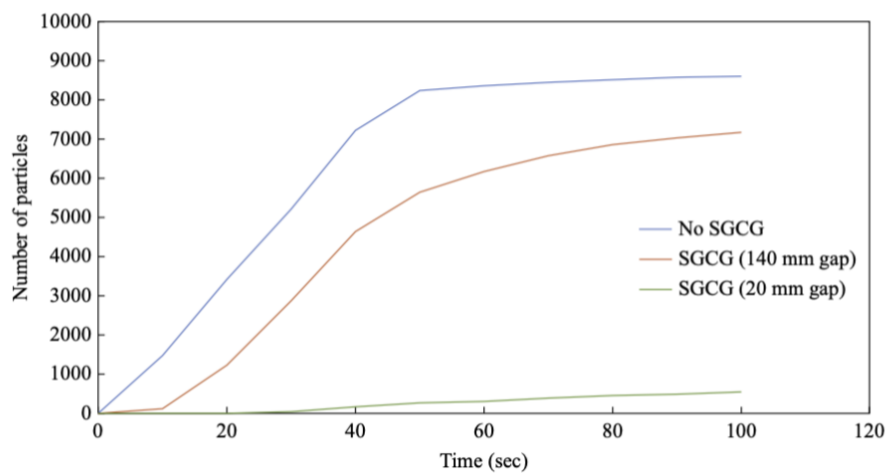
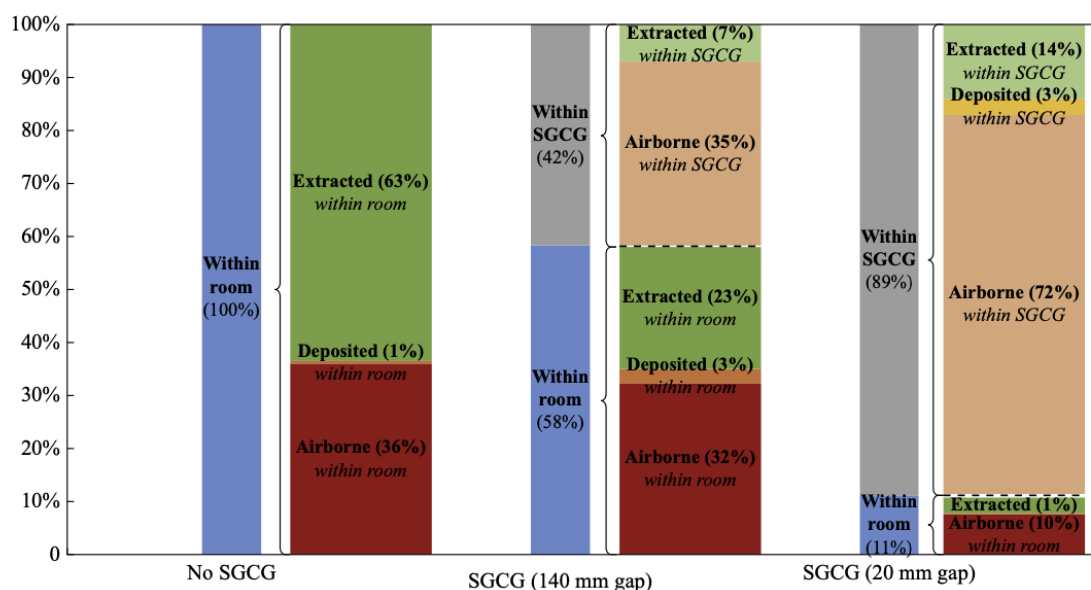


Figure 5. Evolution of the cumulative number of particles passing through the healthcare worker's breathing zone over time depending on the *SplashGuard Caregiver* (SGCG) configuration.



A more detailed analysis was performed in order to better understand the local dynamics of particle concentrations within the room depending on the scenario, as showcased in **Figure 6**. In the baseline configuration, after 100 s of physical time, all the emitted particles in the room either remained airborne (36%), were extracted efficiently by the ventilation system (63%) or deposited on surfaces (1%). This shows that the room had good natural ventilation. In the second configuration (SGCG with 140-mm gap), over half of the particles shed by the patient escaped the SGCG and were in the room (58%), which confirms the previous observations that the air flow within the room tends to enter the SGCG through the gap and exit on the other side. Overall, the number of particles remaining airborne in the room (32%) was comparable to baseline, whilst a lower amount was extracted (23% through ventilation and 7% through the SGCG). A large proportion of particles remained airborne within the SGCG (35%). Finally, the third configuration (SGCG with 20-mm gap) showcases an improved situation as only 10% of the emitted particles remained airborne in the room after 100 s. Seventy-two percent of particles remained airborne within the SGCG, and only 14% were extracted by its ventilation mechanism. It is important to note that the accumulation of aerosols within the SGCG can become an issue when it is removed after the AGP, as they will be released in the room's environment. When analyzing the air flow within the the SGCG, it can be hypothesized that increasing the air suction rate of its negative pressure suction system could lead to improved extraction of airborne particles within the SGCG and a lower concentration in the room.

Figure 6. Breakdown of extracted, deposited and airborne particles within the room and the *SplashGuard Caregiver* (SGCG) for each configuration after 100 s of physical time.



Discussion

Despite the use of PPE and the SGCG, HCWs are susceptible to contamination by viruses, given their continuous presence in the environment, as demonstrated by the viral detection analysis developed, and regardless of the use or not of the SGCG. From this, the aerosolization simulation of the airborne particles aimed to delineate the spread of the virus in the room and identify measures that could be effective in reducing the risk of contamination. This showed that: (1) reducing the gap between the plastic film and the Plexiglas box may decrease the number of particles remaining in the room air significantly (70% less after 100 s); and (2) depending on the room architecture and ventilation design, the position of the HCW could be important in reducing exposure to airborne pathogens.

Despite the adaptations developed in the SGCG, the presence of particles in the environment was observed by viral analysis and by simulation with a common SGCG configuration (gap of 140 mm). From the 3D simulation, it was observed that in the presence of an SGCG with a 140-mm gap, over half (58%) of the particles shed by the patient escape the SGCG and go into the room. In comparison, with a gap of 20 mm, only 10% of particles go into the room air and 72% remain airborne inside the SGCG, showing that reducing the gap seems to be important to improve the protective efficiency of the SGCG. It should also be emphasized that there is a high concentration of particles inside the SGCG, so their removal/movement from the room must be done carefully to avoid spreading them in room air.

CFD simulation has been widely employed as a fast, reliable and cost-effective technique to support decision making and predict mitigation protocols. Its combination with clinical data has optimal applicability, and this was proposed in this study. Physical measurements (phage or gas tracer) to correlate with the CFD analysis were not performed and would add value. Based on this, it was possible to verify that the traditional positioning of the HCW, facing the door and therefore facing the air stream, could expose them to a larger amount of virus and possible contamination. However, with the HCW's back turned towards the door, this exposure could be reduced, even if this higher exposure was not reflected in contamination levels through the sampling, as none of the HCWs evaluated were infected by SARS-CoV-2. This is due to the fact that it is not only the presence of airborne virus which determines infection. The use of PPE, the type of AGP performed (e.g. intubation and positive pressure ventilation seem to be riskier), the duration of exposure of the HCW to the pathogen, their immune status, the clinical condition of the patient and the virulence of the strain can also have an impact [28e30].

Over the course of the pandemic, the science around the modes of transmission of SARS-CoV-2 has evolved, with increasing recognition that transmission occurs through multiple modes of contact with particles of varying sizes. However, the relative contribution of each mode of transmission, and how it may vary according to the environment and circumstances is not well delineated [29,31]. In the present study, the use of PPE probably explains the lack of contamination of the HCWs. However, at the beginning of the pandemic, shortages in PPE were observed at some centers, and HCW positioning combined with SGCG use could have been an option to limit HCW exposure to viral particles. It is important to note that patients' needs were assessed carefully before the device was implemented, and a user manual was drawn up to prevent any misuse, including management of the device in emergency situations (available at <https://rsr-qc.ca/Splashguard-cg/> with English and French versions). No side effects were recorded during the use of the device.

The COVID-19 pandemic resulted in the implementation of several public health and social measures to mitigate spread, which have likely contributed to the significant reduction in seasonal respiratory viruses such as influenza and respiratory syncytial virus in Canada [32], the USA [28] and Europe [33]. Therefore, while these measures were evaluated in the context of SARS-CoV-2, they are certainly transposable to other etiological agents. For example, since the decline of the SARS-CoV-2 pandemic, there has been a major increase in respiratory syncytial infection in paediatrics, and use of the same virus containment measures adopted during the COVID-2 pandemic could be explored [34,35].

The strengths of this study include its prospective design and a pragmatic approach to viral spread, from detection of the virus by accurate methods to the simulation of aerosolized particles within a patient room. In addition, the use of a single room for both virus collection and simulation, addressing all details of its architecture and ventilation, helped to reduce evaluation biases.

Limitations of this study include the single-centre design, the lack of characterization of the strain's virulence, and the small number of patients and AGPs included. Furthermore, based on the viral PCR results, as well as the CFD simulation method (experimental), it was not fully demonstrated that the SGCG offered additional protection over standard PPE. However, by adjusting the device in the CFD (narrowing the gap), the authors were able to identify recommendations for making better use of the SGCG.

In conclusion, this study documented the presence of SARS-CoV-2 in the environment of infected patients without HCW contamination. This indicates that other variables are associated with contamination, as well as the presence of the virus in the air.

Furthermore, the simulation showed that reducing the gap between the plastic film and the Plexiglas box, and modifying the suction port design would make the SGCG more effective. Finally, the position of the HCW within the room can have an impact on their exposure to airborne pathogens.

References

- [1] World Health Organization. Weekly epidemiological update on COVID-19 e 1 June 2023. Geneva: WHO; 2023.
- [2] van Doremalen N, Bushmaker T, Morris DH, Holbrook MG, Gamble A, Williamson BN, et al. Aerosol and surface stability of SARS-CoV-2 as compared with SARS-CoV-1. *N Engl J Med* 2020;382:1564e7.
- [3] Santarpia JL, Rivera DN, Herrera VL, Morwitzer MJ, Creager HM, Santarpia GW, et al. Aerosol and surface contamination of SARS- CoV-2 observed in quarantine and isolation care. *Sci Rep* 2020;10:12732.
- [4] Sepkowitz KA, Eisenberg L. Occupational deaths among health- care workers. *Emerg Infect Dis* 2005;11:1003e8.
- [5] Verbeek JH, Rajamaki B, Ijaz S, Sauni R, Toomey E, Blackwood B, et al. Personal protective equipment for preventing highly infectious diseases due to exposure to contaminated body fluids in healthcare staff. *Cochrane Database Syst Rev* 2020;4:CD011621.
- [6] Nguyen LH, Drew DA, Graham MS, Joshi AD, Guo CG, Ma W, et al. Risk of COVID-19 among front-line health-care workers and the general community: a prospective cohort study. *Lancet Public Heal* 2020;5:e475e83.
- [7] World Health Organization. Infection prevention and control of epidemic- and pandemic-prone acute respiratory infections in health care. Geneva: WHO; 2014.
- [8] Otter JA, Donskey C, Yezli S, Douthwaite S, Goldenberg SD, Weber DJ. Transmission of SARS and MERS coronaviruses and influenza virus in healthcare settings: the possible role of dry surface contamination. *J Hosp Infect* 2016;92:235e50.
- [9] Wilson J, Carson G, Fitzgerald S, Llewelyn MJ, Jenkins D, Parker S, et al. Are medical procedures that induce coughing or involve respiratory suctioning associated with increased generation of aerosols and risk of SARS-CoV-2 infection? A rapid systematic review. *J Hosp Infect* 2021;116:37e46.
- [10] Randall K, Ewing ET, Marr LC, Jimenez JL, Bourouiba L. How did we get here: what are droplets and aerosols and how far do they go? A historical perspective on the transmission of respiratory infectious diseases. *Interface Focus* 2021;11:20210049.

- [11] Peng X, Xu X, Li Y, Cheng L, Zhou X, Ren B. Transmission routes of 2019-nCoV and controls in dental practice. *Int J Oral Sci* 2020;12:9.
- [12] Chang D, Xu H, Rebaza A, Sharma L, Dela Cruz CS. Protecting health-care workers from subclinical coronavirus infection. *Lancet Respir Med* 2020;8:e13.
- [13] Tran K, Cimon K, Severn M, Pessoa-Silva CL, Conly J. Aerosol generating procedures and risk of transmission of acute respiratory infections to healthcare workers: a systematic Review. *PLoS One* 2012;7:e35797.
- [14] Oranger M, Gonzalez-Bermejo J, Dacosta-Noble P, Llontop C, Guerder A, Trosini-Desert V. Continuous positive airway pressure to avoid intubation in SARS-CoV-2 pneumonia: a two-period retrospective case-control study. *Eur Respir J* 2020;56:8e11.
- [15] Simpson JP, Wong DN, Verco L, Carter R, Dzidowski M, Chan PY. Measurement of airborne particle exposure during simulated tracheal intubation using various proposed aerosol containment devices during the COVID-19 pandemic. *Anaesthesia* 2020;75:1587e95.
- [16] Shrimpton AJ, Brown JM, Cook TM, Penfold CM, Reid JP, Pickering AE. Quantitative evaluation of aerosol generation from upper airway suctioning assessed during tracheal intubation and extubation sequences in anaesthetized patients. *J Hosp Infect* 2022;124:13e21.
- [17] Lai HY. Aerosol Box protects healthcare providers during endotracheal intubation.
- [18] Francois T, Tabone L, Levy A, Seguin LA, Toure' T, Aubin CE, et al. Simulation-based rapid development and implementation of a novel barrier enclosure for use in COVID-19 patients: the SplashGuard CG. *Crit Care Res Pract* 2020;2020:3842506.
- [19] Juvet P, Aubin CE. Pare-Partoux SplashGuard-CG. Quebec Respiratory Health Network; 2020. Available at: <https://rsr-qc.ca/Splashguard-cg>
- [20] Canelli R, Connor CW, Gonzalez M, Nozari A, Ortega R. Barrier enclosure during endotracheal intubation. *N Engl J Med* 2020;382:1957e8.
- [21] Buyck M, Levy A, Tabone L, Aubin CE, Juvet P, Baudin F. Pediatric barrier enclosure for nasopharyngeal suctioning during COVID-19 pandemic: a simulation based-study. *Am J Emerg Med* 2021;45:608e10.
- [22] Crawford C, Vanoli E, Decorde B, Lancelot M, Duprat C, Josserand C, et al. Modeling of aerosol transmission of airborne pathogens in ICU rooms of COVID-19 patients with acute respiratory failure. *Sci Rep* 2021;11:11778.
- [23] Chen S, Doolen GD. Lattice Boltzmann method for fluid flows. *Annu Rev Fluid Mech* 1998;30:329e64.

- [24] Dumont-Leblond N, Veillette M, Mubareka S, Yip L, Longtin Y, Jouvét P, et al. Low incidence of airborne SARS-CoV-2 in acute care hospital rooms with optimized ventilation. *Emerg Microbes Infect* 2020;9:2597e605.
- [25] Dumont-Leblond N, Veillette M, Bherer L, Boissoneault K, Mubareka S, Yip L, et al. Positive no-touch surfaces and undetectable SARS-CoV-2 aerosols in long-term care facilities: an attempt to understand the contributing factors and the importance of timing in air sampling campaigns. *Am J Infect Control* 2021;49:701e6.
- [26] Gupta JK, Lin CH, Chen Q. Flow dynamics and characterization of a cough. *Indoor Air* 2009;19:517e25.
- [27] Anand S, Mayya YS. Size distribution of virus laden droplets from expiratory ejecta of infected subjects. *Sci Rep* 2020;10:21174.
- [28] Olsen SJ, Winn AK, Budd AP, Prill MM, Steel J, Midgley CM, et al. Changes in influenza and other respiratory virus activity during the COVID-19 pandemic in the United States, 2020-2021. *MMWR Morb Mortal Wkly Rep* 2021;70:1013e9.
- [29] Shiu EYC, Leung NHL, Cowling BJ. Controversy around airborne versus droplet transmission of respiratory viruses. *Curr Opin Infect Dis* 2019;32:372e9.
- [30] Leal J, Farkas B, Mastikhina L, Flanagan J, Skidmore B, Salmon C, et al. Risk of transmission of respiratory viruses during aerosol-generating medical procedures (AGMPs) revisited in the COVID-19 pandemic: a systematic review. *Antimicrob Resist Infect Control* 2022;11:102.
- [31] Hamilton F, Arnold D, Bzdek BR, Dodd J, AERATOR group, Reid J, et al. Aerosol generating procedures: are they of relevance for transmission of SARS-CoV-2? *Lancet Respir Med* 2021;9:687e9.
- [32] Groves HE, Piché-Renaud PP, Pécia A, Farrar DS, Buckrell S, Bancej C, et al. The impact of the COVID-19 pandemic on influenza, respiratory syncytial virus, and other seasonal respiratory virus circulation in Canada: a population-based study. *Lancet Reg Health Am* 2021;1:100015.
- [33] van Summeren J, Meijer A, Aspelund G, Casalegno JS, Erna G, Hoang U, et al. Low levels of respiratory syncytial virus activity in Europe during the 2020/21 season: what can we expect in the coming summer and autumn/winter? *Eurosurveillance* 2021;26:2100639.
- [34] European Centre for Disease Prevention and Control. Intensified circulation of respiratory syncytial virus (RSV) and associated hospital burden in the EU/EEA. Stockholm: ECDC; 2022.
- [35] Treggiari D, Piubelli C, Formenti F, Silva R, Perandin F. Resurgence of respiratory virus after relaxation of COVID-19 containment measures: a real-world data study from a regional hospital of Italy. *Int J Microbiol* 2022;2022:1e5.

7 CONCLUSÕES

Esta tese demonstrou que a integração entre a mecânica respiratória e a biossegurança é essencial para o manejo das disfunções respiratórias agudas em UTIP. Os dados obtidos neste estudo nos permitem concluir que:

1. A caracterização da mecânica respiratória em lactentes com disfunção respiratória aguda grave evidenciou elevada resistência das vias aéreas e redução moderada da complacência, configurando um padrão fisiopatológico predominantemente obstrutivo;
2. A discrepância entre PIP e PPLAT mostrou-se diretamente relacionada ao aumento da resistência do sistema respiratório e às constantes de tempo prolongadas, evidenciando limitações no uso isolado da PIP para ajuste ventilatório;
3. O modelo experimental *in vitro* apresentou baixa acurácia clínica para estimar a PPLAT a partir da PIP em lactentes com infecção viral grave, não sendo confiável para aplicação isolada à beira-leito;
4. O uso do dispositivo SGCG associou-se à redução da dispersão de partículas virais e da exposição ocupacional durante PGA em ambiente de terapia intensiva;
5. As simulações por CFD demonstraram que a vedação adequada do dispositivo e o posicionamento dos profissionais em na direção oposta ao fluxo principal de ar ambiente reduzem significativamente a concentração de aerossóis no ambiente e;
6. A integração dos achados clínicos e das simulações permitiu propor um modelo prático de manejo ventilatório que concilia a oferta de tratamento ao paciente e a segurança da equipe multiprofissional.

8 CONSIDERAÇÕES FINAIS

Os resultados apresentados ampliam o campo de investigação e apontam para desdobramentos relevantes em pesquisas futuras, incluindo:

1. A avaliação do impacto – em termos de tempo de VM ou morbimortalidade – da aplicação da mecânica respiratória em diferentes cenários pediátricos como uma ferramenta de apoio à tomada de decisão clínica relacionada ao desmame ventilatório ou à modificação de estratégias ventilatórias;
2. A avaliação da mecânica respiratória em pacientes em respiração espontânea, com ênfase na mensuração do esforço inspiratório, atualmente reconhecido como importante determinante para VILI e;
3. Continuidade na investigação de dispositivos de contenção de aerossóis, bem como o aprofundamento das análises aerodinâmicas em ambientes hospitalares.

Tais desdobramentos reforçam o mérito e a relevância deste trabalho, evidenciando seu potencial de continuidade e sua contribuição para o avanço do conhecimento e para a qualificação da assistência em UTIP.

8.1 Framework conceitual

Com base nos achados desta tese e na literatura sobre mecânica respiratória e controle de infecção em unidades de terapia intensiva, propõe-se um *framework* conceitual que integra avaliação fisiológica do sistema respiratório com estratégias de biossegurança. Esse modelo busca orientar a tomada de decisão clínica em cenários de disfunções respiratórias agudas, conciliando a individualização da ventilação mecânica com a redução do risco de exposição ocupacional. Os dois eixos adotados são: **(1) adequação fisiológica da ventilação**, guiada por parâmetros de mecânica respiratória; e **(2) redução do risco de exposição ocupacional**, guiada por medidas de biossegurança proporcionais ao risco de aerossolização, conforme **ANEXO 2**.

REFERÊNCIAS

AMATO, Marcelo B.P. *et al.* Driving Pressure and Survival in the Acute Respiratory Distress Syndrome. **New England Journal of Medicine**, vol. 372, n. 8, p. 747–755, 2015.

ANDREOLIO, Cinara *et al.* Airway resistance and respiratory compliance in children with acute viral bronchiolitis requiring mechanical ventilation support. **Indian Journal of Critical Care Medicine**, vol. 25, n.1, p. 88–93, 2021.

AOYAMA, Hiroko *et al.* Association of driving pressure with mortality among ventilated patients with acute respiratory distress syndrome: A systematic review and meta-analysis. **Critical Care Medicine**, vol. 46, n. 2, p. 300–306, 2018.

ARETS, Hubertus G.M.; VAN DER ENT, Cornelis K. Measurements of airway mechanics in spontaneously breathing young children. **Paediatric Respiratory Reviews**, vol. 5, n. 1, p. 77–84, 2004.

BANSAL, Amit *et al.* SARS-CoV-2 infection rates and associated risk factors in healthcare workers: systematic review and meta-analysis. **Scientific Reports**, vol. 15, n. 1, p. 4705, 2025.

BEAUCHAMP, Francis-Olivier; THÉRIAULT, Julie; SAUTHIER, Michaël. Tailoring ventilation and respiratory management in pediatric critical care: optimizing care with precision medicine. **Current Opinion in Pediatrics**, vol. 37, n. 3, p. 223–232, 2025.

BHALLA, Anoopindar K. *et al.* Adherence to Lung-Protective Ventilation Principles in Pediatric Acute Respiratory Distress Syndrome: A Pediatric Acute Respiratory Distress Syndrome Incidence and Epidemiology Study*. **Critical Care Medicine**, vol. 49, n. 10, p. 1779–1789, 2021.

BOAS, Heather; REN, Clement L. Pediatric Pulmonary Function Testing. **Clinics in Chest Medicine**, vol. 46, n. 3, p. 543–557, 2025.

BRUNO, Francisco *et al.* The Relevance of Airway Resistance in Children Requiring Mechanical Ventilatory Support. **Pediatric Critical Care Medicine**, vol. 23, n. 10, p. e483–e488, 2022.

BURATTI, Cecília Rotava; PIVA, Jefferson; JOUVET, Philippe. Peak Inspiratory Pressure to Estimate Plateau Pressure in Pressure Controlled Modes: Be Aware of Age and Disease*. **Pediatric Critical Care Medicine**, vol. 23, n. 3, p. 225–226, 2022.

CAO, Dazhe *et al.* Part 10: Adult and Pediatric Special Circumstances of Resuscitation: 2025 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. **Circulation**, vol. 152, n. 16_suppl_2, 2025.

COATES, Bria M. *et al.* Influenza A Virus Infection, Innate Immunity, and Childhood. **JAMA Pediatrics**, vol. 169, n. 10, p. 956, 2015.

COHEN, Sapir *et al.* Burden of viral respiratory infections in the pediatric intensive care unit: age, virus distribution, and the impact of the COVID-19 pandemic. **European Journal of Pediatrics**, vol. 184, n. 1, p. 88, 2024.

CRUCES, Pablo. Pediatric Acute Respiratory Distress Syndrome: Approaches in Mechanical Ventilation. **Pediatric Critical Care Medicine**, vol. 24, n. 2, p. e104–e114, 2023.

CRUCES, Pablo *et al.* Plateau Pressure and Driving Pressure in Volume- and Pressure-Controlled Ventilation: Comparison of Frictional and Viscoelastic Resistive Components in Pediatric Acute Respiratory Distress Syndrome. **Pediatric Critical Care Medicine**, vol. 24, n. 9, p. 750–759, 2023.

DE LUCA, Maia *et al.* Severe viral respiratory infections in the pre-COVID era: A 5-year experience in two pediatric intensive care units in Italy. **Influenza and Other Respiratory Viruses**, vol. 17, n. 1, 2023.

D'ERRICO, S. *et al.* Medication Errors in Pediatrics: Proposals to Improve the Quality and Safety of Care Through Clinical Risk Management. **Frontiers in Medicine**, [s. l.], vol. 8, 2022.

DI CICCIO, Maria *et al.* Structural and functional development in airways throughout childhood: Children are not small adults. **Pediatric Pulmonology**, vol. 56, n. 1, p. 240–251, 2021.

DÍAZ, Franco *et al.* Driving Pressure and Normalized Energy Transmission Calculations in Mechanically Ventilated Children Without Lung Disease and Pediatric Acute Respiratory Distress Syndrome*. **Pediatric Critical Care Medicine**, vol. 22, n. 10, p. 870–878, 2021.

DRESSEN, Svenja *et al.* Persistently high burden of acute respiratory infections requiring hospitalization in German pediatric hospitals, fall/winter 2023–2024. **Infection**, vol. 53, n. 2, p. 717–725, 2025.

DUARTE, Helder Brito *et al.* Prediction of static compliance of the respiratory system based on anthropometric measurements in patients on mechanical ventilation: a cross-sectional pragmatic study. **Respiratory Medicine**, vol. 248, p. 108368, 2025.

EMERIAUD, Guillaume *et al.* Executive Summary of the Second International Guidelines for the Diagnosis and Management of Pediatric Acute Respiratory Distress Syndrome (PALICC-2). **Pediatric Critical Care Medicine**, vol. 24, n. 2, p. 143–168, 2023.

FENNELLY, Kevin P. Particle sizes of infectious aerosols: implications for infection control. **The Lancet Respiratory Medicine**, vol. 8, n. 9, p. 914–924, 2020.

FRANÇOIS, Tine *et al.* Simulation-Based Rapid Development and Implementation of a Novel Barrier Enclosure for Use in COVID-19 Patients: The SplashGuard CG. **Critical Care Research and Practice**, vol. 2020, n. December 2019, 2020.

GATTINONI, Luciano *et al.* The future of mechanical ventilation: Lessons from the present and the past. **Critical Care**, vol. 21, n. 1, p. 1–11, 2017.

GÓMEZ-OCHOA, Sergio Alejandro *et al.* COVID-19 in Health-Care Workers: A Living Systematic Review and Meta-Analysis of Prevalence, Risk Factors, Clinical Characteristics, and Outcomes. **American Journal of Epidemiology**, vol. 190, n. 1, p. 161–175, 2021.

GRINNAN, Daniel C.; TRUWIT, Jonathon Dean. Clinical review: Respiratory mechanics in spontaneous and assisted ventilation. **Critical Care**, vol. 9, n. 5, p. 472–484, 2005.

GUÉRIN, Claude *et al.* Effect of driving pressure on mortality in ARDS patients during lung protective mechanical ventilation in two randomized controlled trials. **Critical Care**, vol. 20, n. 1, p. 1–9, 2016.

HE, Congrong *et al.* Particle and bioaerosol characteristics in a paediatric intensive care unit. **Environment International**, vol. 107, p. 89–99, 2017.

HEINONEN, Santtu *et al.* Infant Immune Response to Respiratory Viral Infections. **Immunology and Allergy Clinics of North America**, vol. 39, n. 3, p. 361–376, 2019.

HESS, D. R. Respiratory Mechanics in Mechanically Ventilated Patients. **Respiratory Care**, vol. 59, n. 11, p. 1773–1794, 2014.

ILIA, Stavroula *et al.* Effect of endotracheal tube size, respiratory system mechanics, and ventilator settings on driving pressure. **Pediatric Critical Care Medicine**, p. E47–E51, 2020.

JONES, Anya C. *et al.* Personalized transcriptomics reveals heterogeneous immunophenotypes in children with viral bronchiolitis. **American Journal of Respiratory and Critical Care Medicine**, vol. 199, n. 12, p. 1537–1549, 2019.

JOUVET, Philippe *et al.* Pediatric Acute Respiratory Distress Syndrome: Consensus Recommendations from the Pediatric Acute Lung Injury Consensus Conference. **Pediatric Critical Care Medicine**, vol. 16, n. 5, p. 428–439, 2015.

JOUVET, Philippe; AUBIN, Carl-Eric. **Pare-Partoux SplashGuard-CG**, [s. d.]. Disponible en: <https://rsr-qc.ca/Splashguard-cg/>.

KAMINSKY, David A.; COCKCROFT, Donald W.; DAVIS, Beth E. Respiratory System Dynamics. **Seminars in Respiratory and Critical Care Medicine**, vol. 44, n. 5, p. 526–537, 2023.

KARA, Yalcin *et al.* Assessing the change in the epidemiology of seasonal respiratory viruses with the onset of the COVID-19 pandemic. **Journal of General Virology**, vol. 105, n. 5, 2024.

KASKINEN, Anu K. *et al.* Ventilator-derived dynamic respiratory system compliance: Comparison with static compliance in children. **Respiratory Physiology & Neurobiology**, vol. 249, p. 32–34, 2018.

KHEMANI, Robinder G. *et al.* Paediatric acute respiratory distress syndrome incidence and epidemiology (PARDIE): an international, observational study. **The Lancet Respiratory Medicine**, vol. 7, n. 2, p. 115–128, 2019.

KILLIEN, Elizabeth Y. *et al.* Epidemiology of Intensive Care Admissions for Children in the US From 2001 to 2019. **JAMA Pediatrics**, vol. 177, n. 5, p. 506, 2023.

KLOMPAS, Michael *et al.* Current Insights Into Respiratory Virus Transmission and Potential Implications for Infection Control Programs. **Annals of Internal Medicine**, vol. 174, n. 12, p. 1710–1718, 2021.

KLOMPAS, M. *et al.* Strategies to prevent ventilator-associated pneumonia, ventilator-associated events, and nonventilator hospital-acquired pneumonia in acute-care hospitals: 2022 Update. **Infection Control & Hospital Epidemiology**, [s. l.], vol. 43, n° 6, p. 687–713, 2022.

KNEYBER, Martin C J *et al.* Understanding clinical and biological heterogeneity to advance precision medicine in paediatric acute respiratory distress syndrome. **The Lancet Respiratory Medicine**, vol. 11, n. 2, p. 197–212, 2023.

LAMBERT, Laura; CULLEY, Fiona J. Innate Immunity to Respiratory Infection in Early Life. **Frontiers in Immunology**, vol. 8, 2017.

LEFEBVRE, Marie-Astrid *et al.* Impact of the COVID-19 pandemic on healthcare-associated viral respiratory infections at a tertiary care pediatric hospital. **American Journal of Infection Control**, vol. 51, n. 8, p. 961–963, 2023.

LÓPEZ-FERNÁNDEZ, Yolanda *et al.* Pediatric acute lung injury epidemiology and natural history study: Incidence and outcome of the acute respiratory distress syndrome in children. **Critical Care Medicine**, vol. 40, n.12, p. 3238–3245, 2012.

LÓPEZ-HERCE CID, J. Respiratory function monitoring II: Compliance, airway resistance, dynamic hyperinflation, pulmonary dead-space, work of breathing. **Anales de Pediatría**, vol. 59, n. 3, p. 278–285, 2003.

MAZUR, Natalie I; CABALLERO, Mauricio T; NUNES, Marta C. Severe respiratory syncytial virus infection in children: burden, management, and emerging therapies. **The Lancet**, vol. 404, n. 10458, p. 1143–1156, 2024.

MONROE, Luke W. *et al.* Preventing spread of aerosolized infectious particles during medical procedures: A lab-based analysis of an inexpensive plastic enclosure. **PLOS ONE**, vol. 17, n. 9, p. e0273194, 2022.

NASA, Prashant *et al.* Infection control in the intensive care unit: expert consensus statements for SARS-CoV-2 using a Delphi method. **The Lancet Infectious Diseases**, vol. 22, n. 3, p. e74–e87, 2022.

O'BRYANT, Shelease C. *et al.* Longitudinal evaluation of pediatric respiratory infections. **Journal of Clinical Virology**, vol. 148, p. 105084, 2022.

PANICO, Flávia F. *et al.* Risk factors for mortality and outcomes in pediatric acute lung injury/acute respiratory distress syndrome. **Pediatric Critical Care Medicine**, vol. 16, n. 7, p. e194–e200, 2015.

PATEL, Bhavesh; THOMAS, Neal J.; YEHYA, Nadir. Agreement Between Peak Inspiratory Pressure in Decelerating-Flow Ventilation and Plateau Pressure in Square-Flow Ventilation in Pediatric Acute Respiratory Distress Syndrome. **Pediatric Critical Care Medicine**, vol. Publish Ah, p. Accepted, 2022.

PERCY, Andrew G. *et al.* Mechanical Power Is Associated With Mortality in Pediatric Acute Respiratory Distress Syndrome. **Pediatric Critical Care Medicine**, vol. 24, n. 7, p. e307–e316,

2023.

PETERSON-CARMICHAEL, Stacey *et al.* An official American thoracic society/european respiratory society workshop report: Evaluation of respiratory mechanics and function in the pediatric and neonatal intensive care units. **Annals of the American Thoracic Society**, vol. 13, n. 2, p. S1–S11, 2016.

SEED, Laura; WILSON, David; COATES, Allan L. Children Should Not Be Treated Like Little Adults in the PFT Lab. **Respiratory Care**, vol. 57, n. 1, p. 61–74, 2012.

SERPA NETO, Ary *et al.* Mechanical power of ventilation is associated with mortality in critically ill patients: an analysis of patients in two observational cohorts. **Intensive Care Medicine**, vol. 44, n. 11, p. 1914–1922, 2018.

SIEGEL, Jane D. *et al.* 2007 Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Health Care Settings. **American Journal of Infection Control**, vol. 35, n. 10, p. S65–S164, 2007.

SMITH, H. A. B. *et al.* 2022 Society of Critical Care Medicine Clinical Practice Guidelines on Prevention and Management of Pain, Agitation, Neuromuscular Blockade, and Delirium in Critically Ill Pediatric Patients With Consideration of the ICU Environment and Early Mobility. **Pediatric Critical Care Medicine**, [s. l.], vol. 23, n° 2, p. e74–e110, 2022.

TEPPER, ROBERT S. *et al.* Static Compliance of the Respiratory System in Healthy Infants. **American Journal of Respiratory and Critical Care Medicine**, vol. 163, n. 1, p. 91–94, 2001.

THURESSON, Sara *et al.* Airborne Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) in Hospitals: Effects of Aerosol-Generating Procedures, HEPA-Filtration Units, Patient Viral Load, and Physical Distance. **Clinical Infectious Diseases**, vol. 75, n. 1, p. e89–e96, 2022.

TIAN, Chenchen *et al.* Risk factors and protective measures for healthcare worker infection during highly infectious viral respiratory epidemics: A systematic review and meta-analysis.

Infection Control & Hospital Epidemiology, vol. 43, n. 5, p. 639–650, 2022.

TRACHSEL, Daniel *et al.* Developmental respiratory physiology. **Pediatric Anesthesia**, vol. 32, n. 2, p. 108–117, 2022.

TRAN, Khai *et al.* Aerosol Generating Procedures and Risk of Transmission of Acute Respiratory Infections to Healthcare Workers: A Systematic Review. **PLoS ONE**, vol. 7, n. 4, p. e35797, 2012.

UYEKI, Timothy M *et al.* Influenza. **The Lancet**, vol. 400, n. 10353, p. 693–706, 2022.

VAN SCHELVEN, Patrick *et al.* Driving Pressure Is Associated With Outcome in Pediatric Acute Respiratory Failure. **Pediatric Critical Care Medicine**, vol. Publish Ah, p. 1–9, 2021.

VERHOEVEN, David. Immunometabolism and innate immunity in the context of immunological maturation and respiratory pathogens in young children. **Journal of Leukocyte Biology**, vol. 106, n. 2, p. 301–308, 2019.

YEHYA, Nadir *et al.* Definition, Incidence, and Epidemiology of Pediatric Acute Respiratory Distress Syndrome: From the Second Pediatric Acute Lung Injury Consensus Conference. **Pediatric Critical Care Medicine**, vol. 24, n. Supplement 1 2S, p. S87–S98, 2023.

YEHYA, Nadir *et al.* Respiratory Dysfunction Criteria in Critically Ill Children: The PODIUM Consensus Conference. **Pediatrics**, vol. 149, n. Supplement_1, p. S48–S52, 2022.

ZANELLA, Marie-Céline; RHEE, Chanu; KLOMPAS, Michael. Preventing respiratory viral transmission in healthcare settings. **Current Opinion in Infectious Diseases**, vol. 38, n. 4, p. 339–346, 2025.

ANEXO A – CERTIFICADO RELATIVO AO FELLOWSHIP DE PESQUISA EM FISIOLOGIA DO SUPORTE VENTILATÓRIO EM CRIANÇAS

Montréal, March 6 2024

To whom is concerned



CHU Sainte-Justine
Le centre hospitalier
universitaire mère-enfant

Pour l'amour des enfants

Université 
de Montréal

Object : Certificate of a 9 months research fellowship of Cecilia Buratti MD MSc, PhD student in the Research Center of Ste-Justine Hospital-Montreal.

Dear Madam or Sir,

I certify that Dr Cecilia Buratti performed a PhD research internship on respiratory support physiology in children in my laboratory from September 15th 2021 to June 15th 2022. Her work was a full time research fellowship of 35 hours a week during this 9 months period.

With my best regards.

Philippe Juvet MD PhD MBA

Pediatric intensivist
Full Professor in Pediatrics and in Public Health
Director of the Respiratory Research Network of Québec
Director of the clinical decision support system lab

CHU Sainte-Justine
3175 Chemin de Côte Sainte Catherine - Montreal (Québec) H3T 1C5 - Canada
Secrétariat (Mr Ludovic Lollia) : (1) 514 345 4931 # 3936
Tel direct : (1) 514 345 4927
E-mail address : philippe.juvet.med@ssss.gouv.qc.ca

3175, Côte-Sainte-Catherine
Montréal (Québec)
H3T 1C5

ANEXO A – *FRAMEWORK* CONCEITUAL: INTEGRANDO MECÂNICA RESPIRATÓRIA E BIOSSEGURANÇA

Manejo Integrado de Disfunções Respiratórias em UTIP: Framework de Fisiologia e Biossegurança



PACIENTE

1 ETAPA 1: RECONHECIMENTO DO CENÁRIO

IDENTIFICAÇÃO DO FENÓTIPO RESPIRATÓRIO



Obstrutiva
(ex: Bronquiolite)
Restritiva
(ex: SDRA)
Mista

Classificar para direcionar ajuste ventilatório inicial.

CLASSIFICAÇÃO DO RISCO DE TRANSMISSIBILIDADE

